MIL-M-28787/34B(EC)
14 January 1983
SUPERSEDING
MIL-M-28787/34A(EC)
28 September 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE JBN

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic monostable multivibrator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.95 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JBN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/34-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

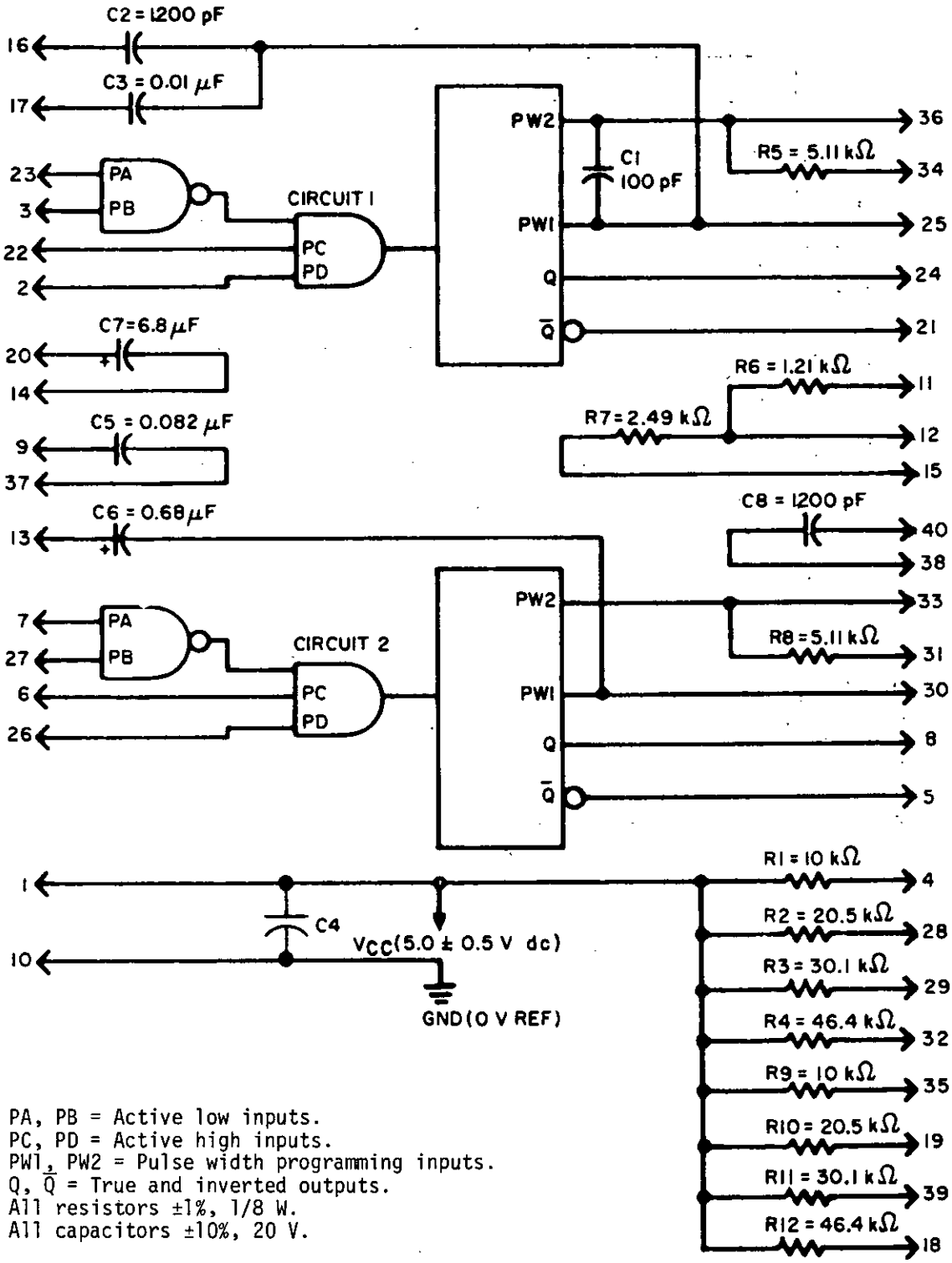
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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PA, PB = Active low inputs.
PC, PD = Active high inputs.
PW1, PW2 = Pulse width programming inputs.
Q, $\overline{Q}$ = True and inverted outputs.
All resistors ±1%, 1/8 W.
All capacitors ±10%, 20 V.

* FIGURE 1. JBN module function diagram, monostable multivibrator.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1-B8	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
C	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D8	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E8	Input current, high level	VII	X	0	40	0	40	0	40	0	40	µA
F1-F8	Input current, low level	VII	X	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	mA
G1-G4	Output short circuit current	VII	X	10	40	10	40	10	40	10	40	mA
H1-H4	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	mV
J1-J4	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K	Power supply current	IX	X	15	50	15	50	15	50	15	50	mA
L1,L2	Resistance	X	X	9,900	10,100	9,800	10,200	9,850	10,150	9,785	10,215	Ω
L3,L4	Resistance	X	X	20,295	20,705	20,090	20,910	20,192	20,808	20,059	20,941	Ω
L5,L6	Resistance	X	X	29,799	30,401	29,498	30,702	29,640	30,560	29,452	30,748	Ω
L7,L8	Resistance	X	X	45,936	46,864	45,472	47,328	45,700	47,100	44,772	48,028	Ω
L9,L10	Resistance	X	X	5,059	5,161	5,008	5,212	5,033	5,187	5,000	5,220	Ω
L11	Resistance	X	X	1,198	1,222	1,186	1,234	1,192	1,228	1,174	1,246	Ω
L12	Resistance	X	X	2,465	2,515	2,430	2,550	2,453	2,527	2,426	2,554	Ω
M1-M6	Capacitor dc leakage current	XI	X	0	3	0	3	0	3	0	3	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1	Capacitance	XII	X	90	130	90	130	80	140	80	140	pF
N2	Capacitance	XII	X	1,080	1,320	1,080	1,320	960	1,440	960	1,440	pF
N3	Capacitance	XII	X	0.009	0.011	0.009	0.011	0.008	0.012	0.008	0.012	µF
N4	Capacitance	XII	X	0.0738	0.0902	0.0738	0.0902	0.0656	0.0984	0.0656	0.0984	µF
N5	Capacitance	XII	X	0.612	0.748	0.612	0.748	0.544	0.816	0.544	0.816	µF
N6	Capacitance	XII	X	6.12	7.48	6.12	7.48	5.44	8.16	5.44	8.16	µF
N7	Capacitance	XII	X	1,080	1,320	1,080	1,320	960	1,440	960	1,440	pF
P1-P4	Capacitor dissipation factor	XII	X	0.0	2.5	0.0	2.5	0.0	2.5	0.0	2.5	%
P5,P6	Capacitor dissipation factor	XII	X	0	3	0	3	0	3	0	3	%
Q1-Q8	Pulse width	XIII	X	1.86	2.98	1.86	2.98	1.62	3.22	1.70	3.14	µs
Q9-Q16	Pulse width	XIII	X	1.72	2.74	1.72	2.74	1.50	2.96	1.50	2.96	µs
Q17,Q18, Q95,Q96	Pulse width	XIII	S	423	677	423	677	359	741	359	741	ns
Q19,Q20, Q97,Q98	Pulse width	XIII	S	847	1,353	847	1,353	721	1,479	721	1,479	ns
Q21,Q22, Q99,Q100	Pulse width	XIII	S	1,155	1,845	1,155	1,845	983	2,017	983	2,017	ns
Q23,Q24, Q101,Q102	Pulse width	XIII	S	1,733	2,787	1,733	2,787	1,470	3,050	1,470	3,050	ns
Q25,Q26, Q93,Q94	Pulse width	XIII	S	223	357	223	357	189	391	189	391	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q27,Q28	Pulse width	XIII	S	3.43	5.47	3.43	5.47	2.92	5.98	2.92	5.98	µs
Q29,Q30	Pulse width	XIII	S	6.79	10.85	6.79	10.85	5.78	11.86	5.78	11.86	µs
Q31,Q32	Pulse width	XIII	S	9.9	15.7	9.9	15.7	8.4	17.2	8.4	17.2	µs
Q33,Q34	Pulse width	XIII	S	15.1	24.1	15.1	24.1	12.9	26.3	12.9	26.3	µs
Q35,Q36	Pulse width	XIII	S	14.5	23.1	14.5	23.1	12.4	25.2	12.4	25.2	µs
Q37,Q38	Pulse width	XIII	S	26.7	42.5	26.7	42.5	22.7	46.5	22.7	46.5	µs
Q39,Q40	Pulse width	XIII	S	52.8	84.2	52.8	84.2	44.9	92.1	44.9	92.1	µs
Q41,Q42	Pulse width	XIII	S	76.7	112.5	76.7	112.5	67.7	121.5	67.7	121.5	µs
Q43,Q44	Pulse width	XIII	S	117.2	187.2	117.2	187.2	99.7	204.7	99.7	204.7	µs
Q45,Q46	Pulse width	XIII	S	9.7	15.5	9.7	15.5	8.2	17.0	8.2	17.0	ms
Q47,Q48	Pulse width	XIII	S	17.9	28.7	17.9	28.7	15.2	31.4	15.2	31.4	ms
Q49,Q50	Pulse width	XIII	S	35.5	56.7	35.5	56.7	30.2	62.0	30.2	62.0	ms
Q51,Q52	Pulse width	XIII	S	51.6	82.4	51.6	82.4	43.9	90.1	43.9	90.1	ms
Q53,Q54	Pulse width	XIII	S	79	126	79	126	66.3	138.7	66.3	138.7	ms
Q55,Q56	Pulse width	XIII	S	46	101	46	101	40	111	40	111	ns
Q57,Q58	Pulse width	XIII	S	116	185	116	185	99	202	99	202	ns
Q59,Q60	Pulse width	XIII	S	216	344	216	344	184	376	184	376	ns
Q61,Q62	Pulse width	XIII	S	308	492	308	492	262	538	262	538	ns
Q63,Q64	Pulse width	XIII	S	477	763	477	763	405	835	405	835	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q65,Q66	Pulse width	XIII	S	3.17	5.07	3.17	5.07	2.70	5.54	2.70	5.54	µs
Q67,Q68	Pulse width	XIII	S	6.27	10.01	6.27	10.01	5.33	10.95	5.33	10.95	µs
Q69,Q70	Pulse width	XIII	S	9.1	14.5	9.1	14.5	7.7	15.9	7.7	15.9	µs
Q71,Q72	Pulse width	XIII	S	13.9	22.3	13.9	22.3	11.8	24.4	11.8	24.4	µs
Q73,Q74	Pulse width	XIII	S	117	187	117	187	100	204	100	204	µs
Q75,Q76	Pulse width	XIII	S	216	346	216	346	184	378	184	378	µs
Q77,Q78	Pulse width	XIII	S	428	684	428	684	364	748	364	748	µs
Q79,Q80	Pulse width	XIII	S	622	994	622	994	528	1,088	528	1,088	µs
Q81,Q82	Pulse width	XIII	S	952	1,520	952	1.520	810	1,662	810	1,662	µs
Q83,Q84	Pulse width	XIII	S	0.97	1.55	0.97	1.55	0.82	1.70	0.82	1.70	ms
Q85,Q86	Pulse width	XIII	S	1.79	2.87	1.79	2.87	1.52	3.14	1.52	3.14	ms
Q87,Q88	Pulse width	XIII	S	3.55	5.67	3.55	5.67	3.02	6.08	3.02	6.08	ms
Q89,Q90	Pulse width	XIII	S	5.16	8.24	5.16	8.24	4.39	9.01	4.39	9.01	ms
Q91,Q92	Pulse width	XIII	S	7.89	12.61	7.89	12.61	6.71	13.79	6.71	13.79	ms
R1-R4	Transition time (t_{TLH})	XIII	S	10	80	10	80	10	85	10	85	ns
S1-S4	Transition time (t_{THL})	XIII	S	2	25	2	25	2	30	2	30	ns
T1-T16	Propagation delay time (t_{PLH})	XIII	S	5	49	5	49	5	53	5	53	ns
U1-U16	Propagation delay time (t_{PHL})	XIII	S	5	49	5	49	5	54	5	54	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V1-V8	Retrigger pulse width	XIV	S	228	406	228	406	199	435	199	435	ns
V9-V16	Retrigger pulse width	XIV	S	1.72	2.74	1.72	2.74	1.50	2.96	1.50	2.96	µs
W	Power supply filter capacitance	XV	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
X1-X4	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
Y1-Y4	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
AA1-AA8	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	pF
BB1,BB2	Stray capacitance	XVIII	Q	0	100	0	100	0	100	0	100	pF
BB3,BB4	Stray capacitance	XVIII	Q	0	50	0	50	0	50	0	50	pF
BB5	Stray capacitance	XVIII	Q	5	30	5	30	5	30	5	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, C, or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, G where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the oerating temperature tests, the module shall be operated in the test circuit shown on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and while still at the maximum class temperature, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military speciation MIL-M-28787/34 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833446. Part number M28787/34-1 supersedes part numbers 2833446 and M28787/34.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	285 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage	
Low level	0 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, low level	
(V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum

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Input current, high level	
(V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Output voltage	
High level	2.4 V minimum, V_{CC} maximum
Low level	0 V minimum, 0.4 V maximum
Stray capacitance	50pF maximum
Timing resistor (range)	5 kΩ minimum, 50 kΩ maximum
Output drive current, high level, maximum capability	
(V_{OH} = 2.4 V)	-0.72 mA
Output drive current, low level, maximum capability	
(V_{OL} = 0.4 V)	9.6 mA
Propagation delay time	
t_{PLH}	5 ns minimum, 53 ns maximum
t_{PHL}	5 ns minimum, 54 ns maximum
Input setup time	40 ns minimum
Input hold time	40 ns minimum

* 6.5 Module function. This module contains two retriggerable monostable multivibrator circuits and a range of discrete resistors and capacitors. Each circuit has four inputs: two active HIGH and two active LOW. This allows a choice of triggering on the leading or trailing edge of the input waveform. Each monostable multivibrator circuit has both true and inverted outputs with the output pulse width being pin programmable. When the input trigger conditions are met, each monostable multivibrator circuit generates one output pulse.

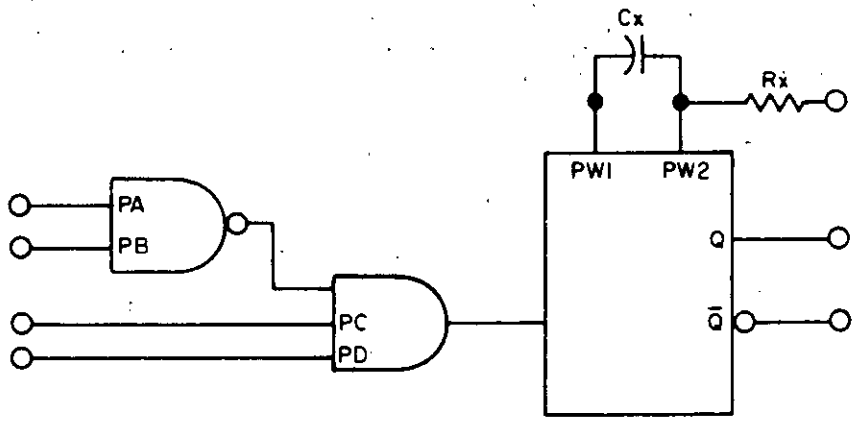
6.5.1 Input to output relationship. Each retriggerable monostable multivibrator circuit has the input to output relationship of table II and figure 2.

TABLE II. Input to output relationship.

PA	PB	PC	PD	Output operation
H→L	H	H	H	Trigger
H	H→L	H	H	Trigger
L	X	L→H	H	Trigger
X	L	L→H	H	Trigger
L	X	H	L→H	Trigger
X	L	H	L→H	Trigger

H = High level.
L = Low level.
X = Don't care condition.
H→L = Transition from high to low level.
L→H = Transition from low to high level.

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PIN DESIGNATIONS:

PA, PB	= Active low inputs.	$\overline{Q}$	= Inverted output.
PC, PD	= Active high inputs.	Cx	= External capacitance.
Q	= True output.	Rx	= External timing resistor.

FIGURE 2. Input to output relationship of each retriggerable monostable multivibrator circuit.

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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Rules for operation. The following list of rules are recommended for proper operation of each monostable multivibrator.

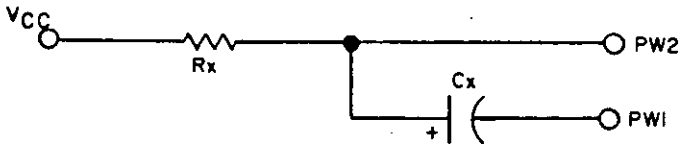
a. An external resistor (Rx) and external capacitor (Cx) are shown on figure 3.
b. The value of resistor Rx may vary from 5.0 kΩ to 50 kΩ for 0°C to 60°C operation.
c. Capacitor Cx may vary from zero to any necessary value available.
d. If capacitor Cx has leakages approaching 3.0 µA, or if stray capacitance from either terminal to ground is more than 50 pF, the timing equations may not represent the pulse width obtained.
e. The output pulse width (t) is defined as follows:

For $Cx \geq 1{,}000$ pF, $t = 0.32\text{-}RxCx\left(1 + \frac{0.7}{Rx}\right)$

where Rx is in kΩ, Cx is in pF, and t is in ns.

For Cx < 1,000 pF, t can be obtained from figure 3.

f. If electrolytic type capacitors are to be used, the following three configurations are recommended:
(1) For use with low leakage electrolytic capacitors:

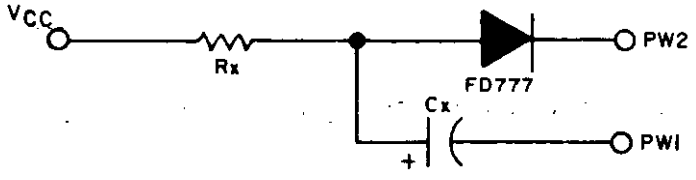


t = see rule (e)

The normal RC configuration can be used predictably if the forward capacitor leakage at 5.0 V is less than 3 µA, and the inverse capacitor leakage at 1.0 V is less than 5 µA over the operational temperature range.

(2) For use with high inverse leakage current electrolytic capacitors:

R < 0.6 Rx (maximum)

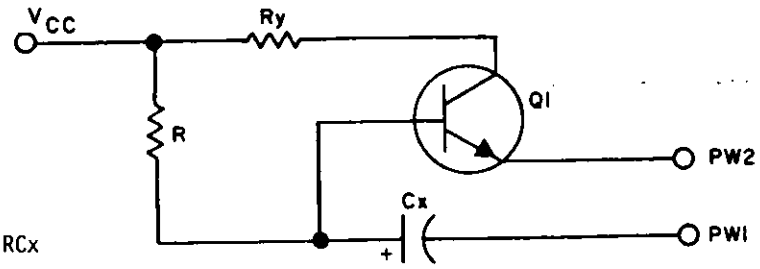


t ≈ 0.3 RCx

The diode in this configuration prevents high inverse leakage currents through the capacitor by preventing an inverse voltage across the capacitor.

(3) Use to obtain extended pulse widths:

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$t \approx 0.3\ RCx$

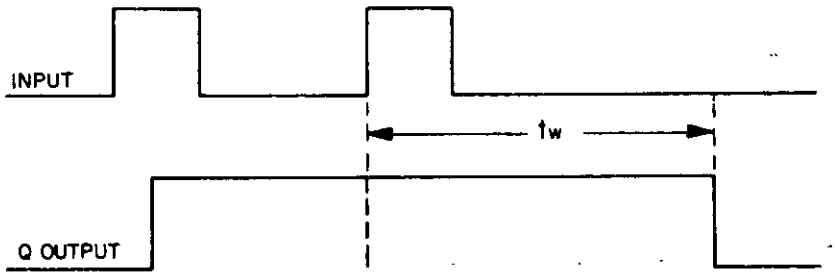
This configuration obtains extended pulse widths because of the larger timing resistor allowed by Beta multiplication. Electrolytics with high inverse leakage currents can be used.

R < Rx (0.7) (hFE Q1) or < 2.5 MΩ, whichever is less.
Rx (minimum) < Ry < Rx (maximum) (5 ≤ Ry ≤ 10 kΩ is recommended).
Q1: NPN silicon transistor with hFE requirements of above equations, such as 2N5961 or 2N5962.

Configuration (2) and (3) are not recommended with retriggerable operation.

g. Under any operating condition, Cx and Rx (minimum) must be kept as close to the circuit as possible to minimize stray capacitance and reduce noise pickup.

h. The retrigger pulse width is calculated as shown below:



$$tw = t + tPD\text{-}1 = 0.32\ RxCx\ (1 + \frac{0.7}{Rx}) + tPD\text{-}1$$

The retrigger pulse width is equal to the pulse width t plus a delay time. For pulse widths greater than 500 ns, tw can be approximated as t.

NOTE: Retriggering will not occur if the retrigger pulse comes within ≈0.3 Cx ns after the initial trigger pulse, that is, during the discharge cycle time.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N408)

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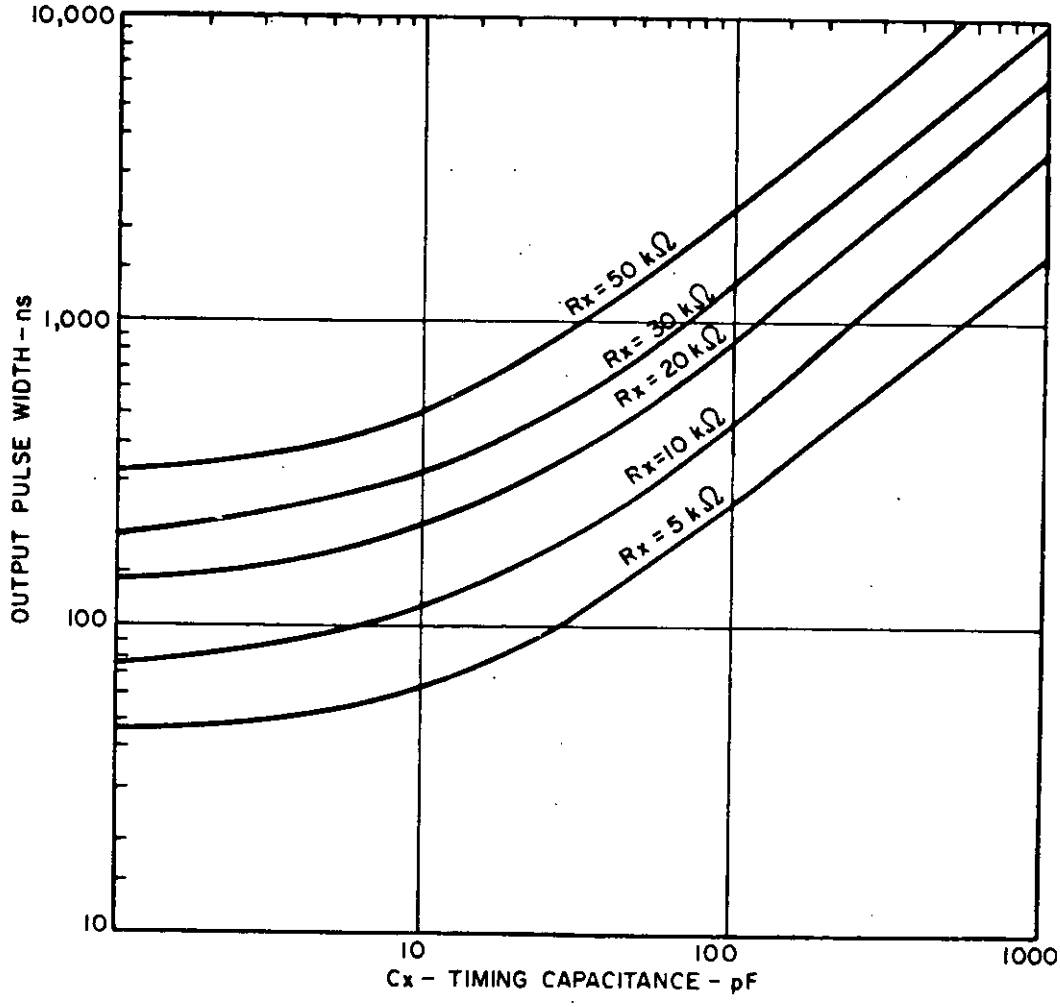
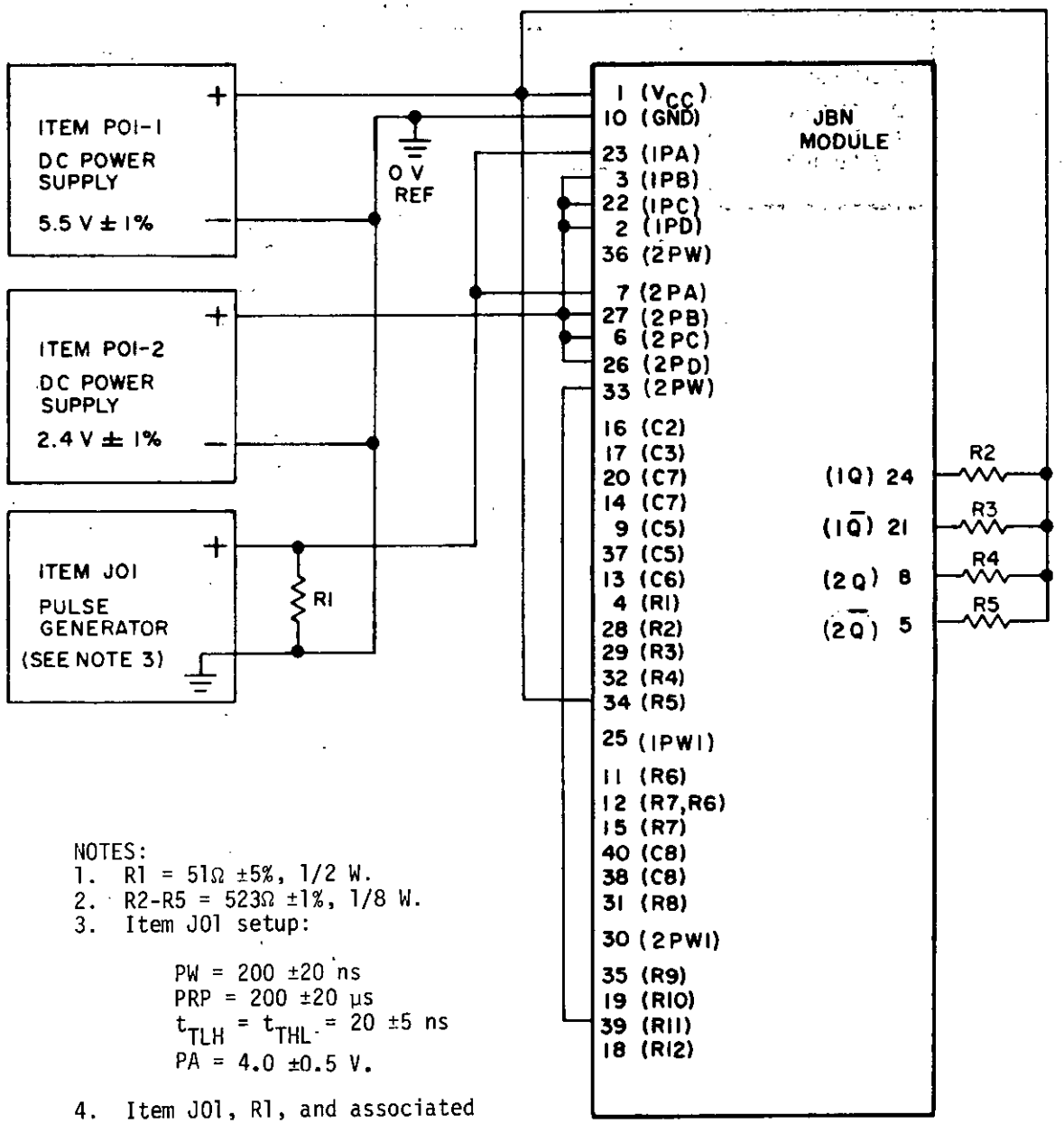


FIGURE 3. Graph of timing capacitance versus output pulse width.

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NOTES:
1. RI = 51Ω ±5%, 1/2 W.
2. R2-R5 = 523Ω ±1%, 1/8 W.
3. Item JOI setup:

   PW = 200 ±20 ns
   PRP = 200 ±20 µs
   $t_{TLH} = t_{THL}$ = 20 ±5 ns
   PA = 4.0 ±0.5 V.

4. Item JOI, RI, and associated wiring is not used for component temperature test.

* FIGURE 4. Operating temperature, component temperature, and life test circuit.

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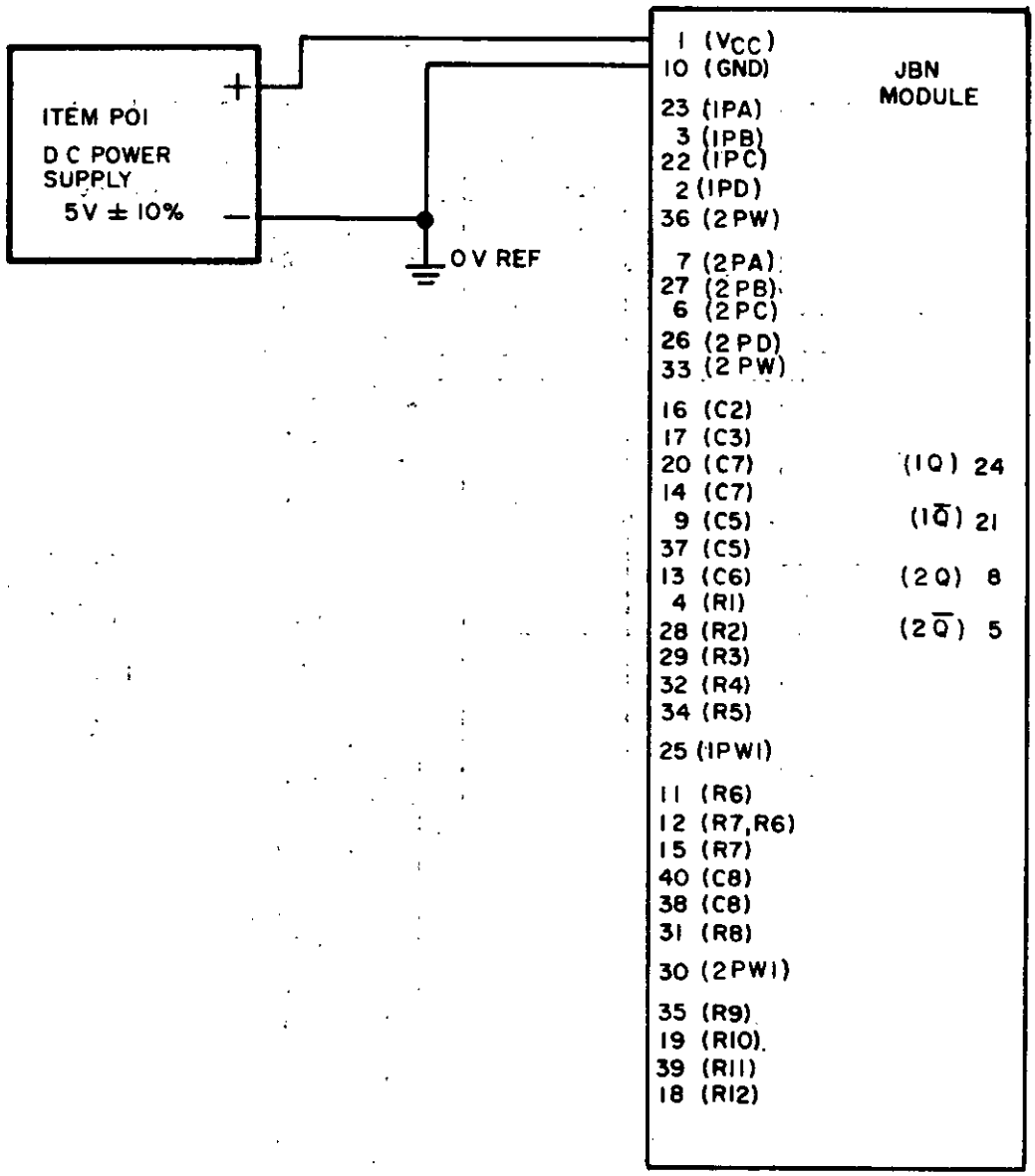


FIGURE 5. Vibration test circuit.

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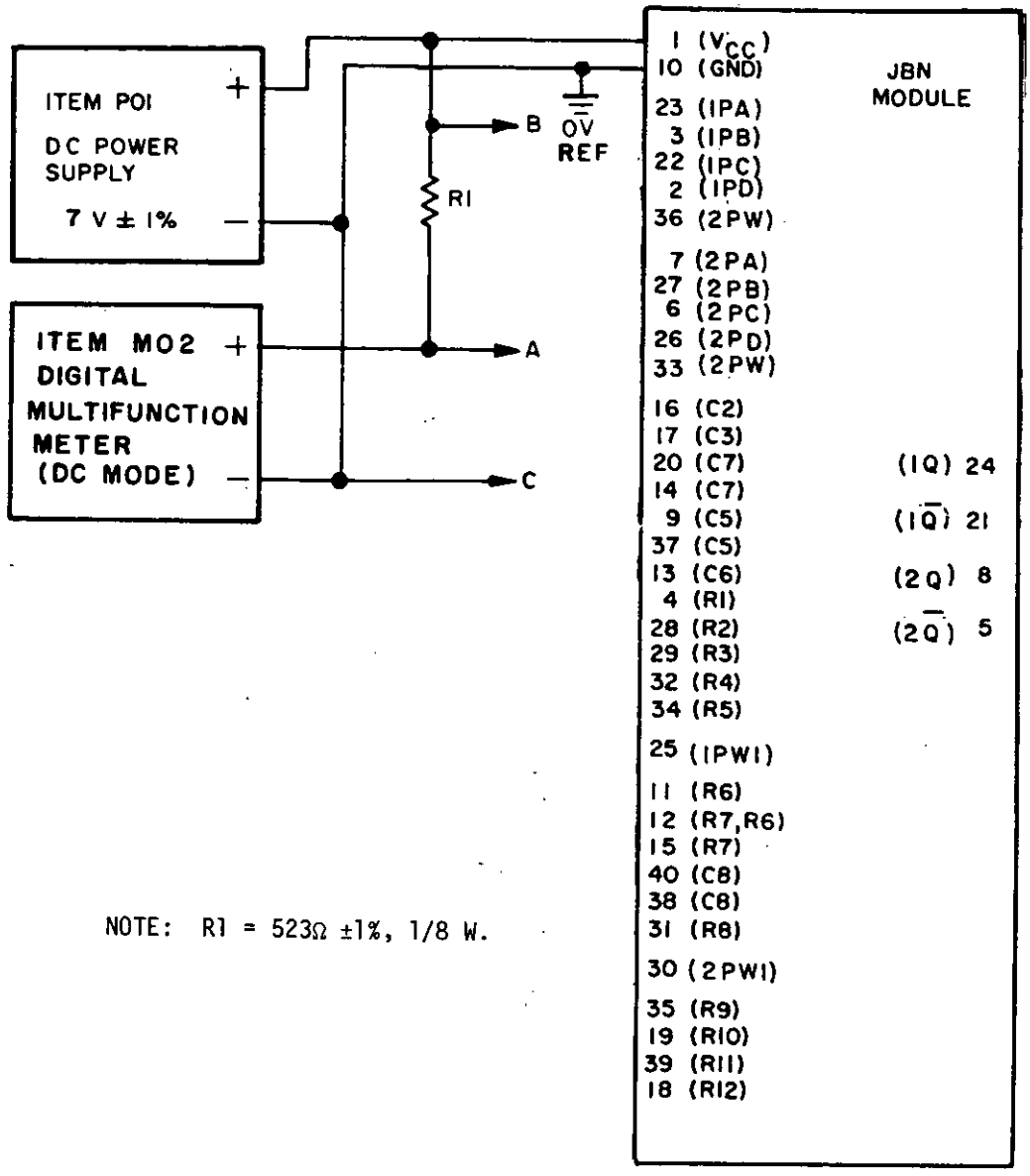
* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
A1	Q1	V_{OUT} (low loaded)	24	34	Open	10	800	mV	6
A2	$\overline{Q1}$	"	21	34	25	"	"	"	"
A3	Q2	"	8	31	Open	"	"	"	"
A4	$\overline{Q2}$	"	5	31	30	"	"	"	"

TABLE IV. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1PA	Input breakdown current	3	23	0	1	mA	7
B2	1PB	"	23	3	"	"	"	"
B3	1PC	"	2	22	"	"	"	"
B4	1PD	"	22	2	"	"	"	"
B5	2PA	"	27	7	"	"	"	"
B6	2PB	"	7	27	"	"	"	"
B7	2PC	"	26	6	"	"	"	"
B8	2PD	"	6	26	"	"	"	"

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NOTE: RI = 523Ω ±1%, 1/8 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

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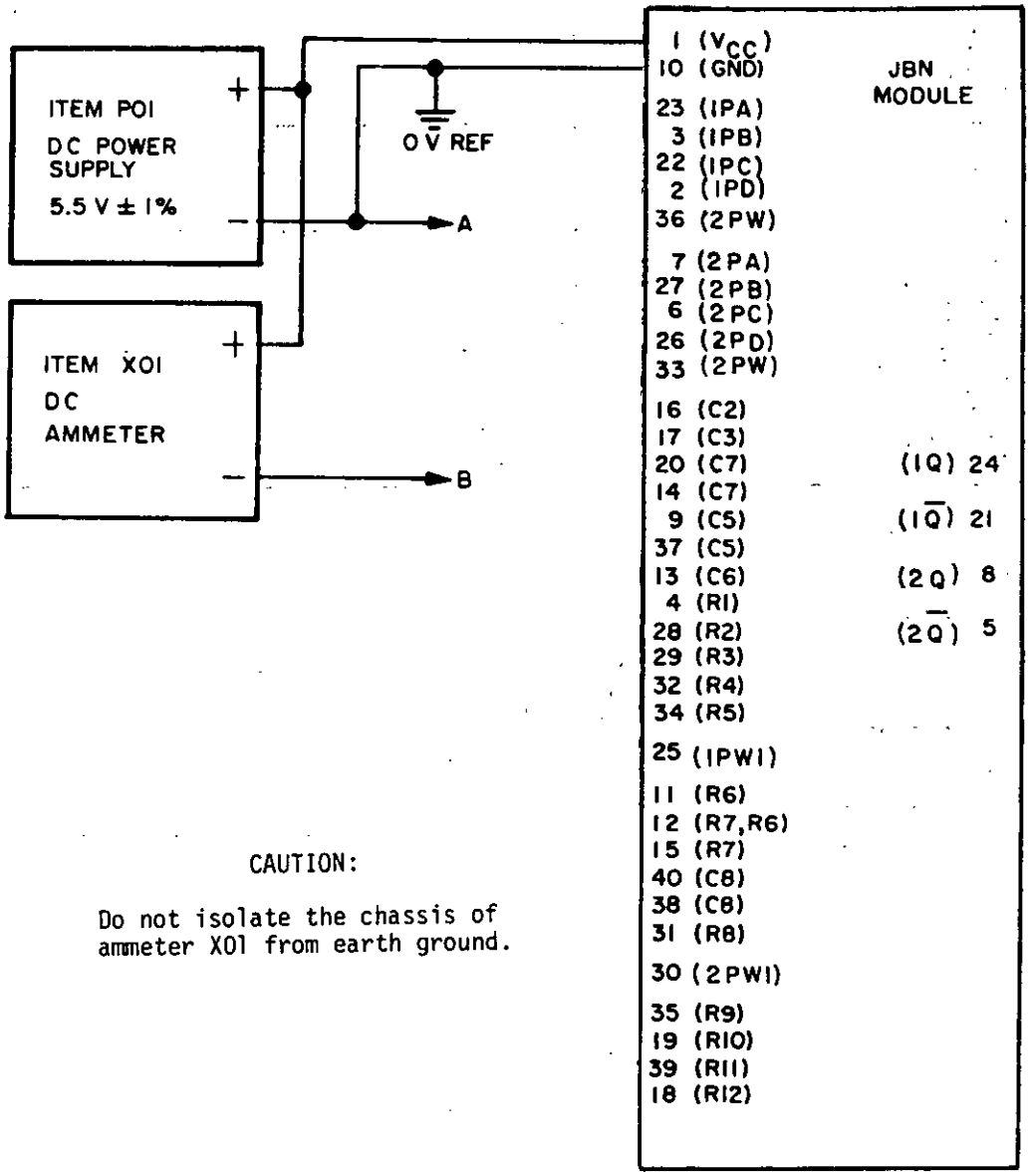


FIGURE 7. Input breakdown current test circuit for table IV.

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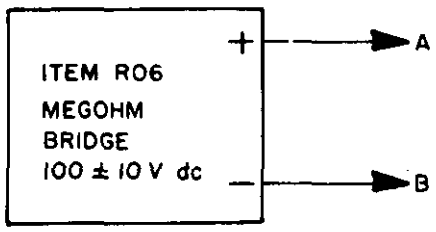
TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	All pins	Frame	10	∞	MΩ	8

TABLE VI. Input clamp diode voltage test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	1PA	Input clamp diode voltage	23	-0.4	-1.5	V	9
D2	1PB	"	3	"	"	"	"
D3	1PC	"	22	"	"	"	"
D4	1PD	"	2	"	"	"	"
D5	2PA	"	7	"	"	"	"
D6	2PB	"	27	"	"	"	"
D7	2PC	"	6	"	"	"	"
D8	2PD	"	26	"	"	"	"

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WARNING:

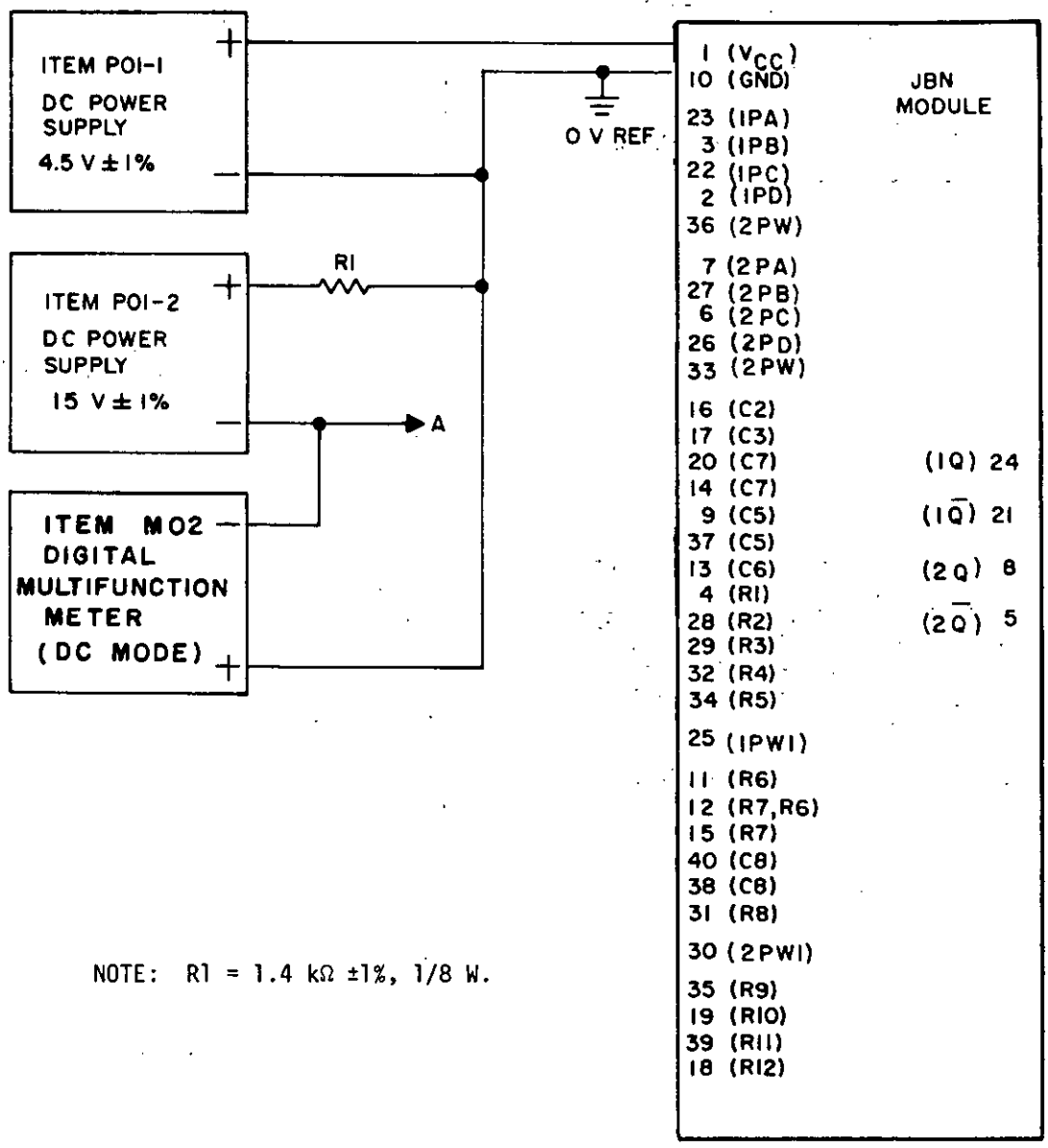
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

JBN MODULE	
1 (V_{CC})	
10 (GND)	
23 (1PA)	
3 (1PB)	
22 (1PC)	
2 (1PD)	
36 (2PW)	
7 (2PA)	
27 (2PB)	
6 (2PC)	
26 (2PD)	
33 (2PW)	
16 (C2)	
17 (C3)	
20 (C7)	(1Q) 24
14 (C7)	
9 (C5)	($1\overline{Q}$) 21
37 (C5)	
13 (C6)	(2Q) 8
4 (R1)	
28 (R2)	($2\overline{Q}$) 5
29 (R3)	
32 (R4)	
34 (R5)	
25 (1PW1)	
11 (R6)	
12 (R7,R6)	
15 (R7)	
40 (C8)	
38 (C8)	
31 (R8)	
30 (2PW1)	
35 (R9)	
19 (R10)	
39 (R11)	
18 (R12)	

FIGURE 8. Isolation test circuit for table V.

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NOTE: R1 = 1.4 kΩ ±1%, 1/8 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (high-level and low-level) and output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1PA Input current, high level	23	Open	3	Open	0	40	µA	10
E2	1PB "	3	"	23	"	"	"	"	"
E3	1PC "	22	"	2	"	"	"	"	"
E4	1PD "	2	"	22	"	"	"	"	"
E5	2PA "	7	"	27	"	"	"	"	"
E6	2PB "	27	"	7	"	"	"	"	"
E7	2PC "	6	"	26	"	"	"	"	"
E8	2PD "	26	"	6	"	"	"	"	"
F1	1PA Input current, low level	Open	23	Open	3	0.10	1.73	mA	"
F2	1PB "	"	3	"	23	"	"	"	"
F3	1PC "	"	22	"	2	"	"	"	"
F4	1PD "	"	2	"	22	"	"	"	"
F5	2PA "	"	7	"	27	"	"	"	"
F6	2PB "	"	27	"	7	"	"	"	"
F7	2PC "	"	6	"	26	"	"	"	"
F8	2PD "	"	26	"	6	"	"	"	"

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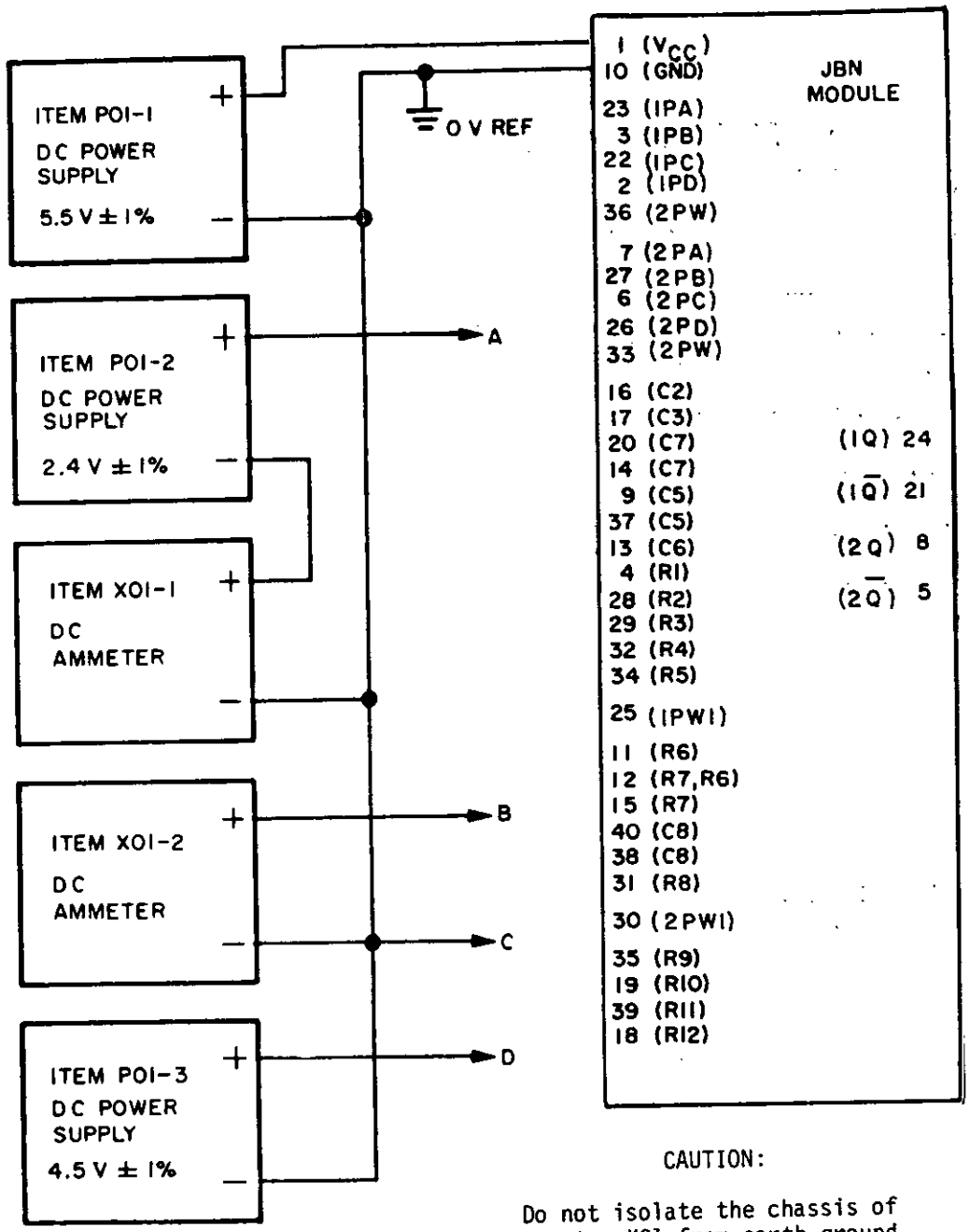
TABLE VII. Input current (high-level and low-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q	Output short circuit current	Open	24	25	34	10	40	mA	10
G2	$1\overline{Q}$	"	"	21	Open	34	"	"	"	"
G3	2Q	"	"	8	30	31	"	"	"	"
G4	$2\overline{Q}$	"	"	5	Open	31	"	"	"	"

* TABLE VIII. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H1	1Q	Output voltage, low level	24	Open	0	400	mV	11
H2	$1\overline{Q}$	"	21	25	"	"	"	"
H3	2Q	"	8	Open	"	"	"	"
H4	$2\overline{Q}$	"	5	30	"	"	"	"
J1	1Q	Output voltage, high level	24	25	2.4	4.5	"	"
J2	$1\overline{Q}$	"	21	Open	"	"	"	"
J3	2Q	"	8	30	"	"	"	"
J4	$2\overline{Q}$	"	5	Open	"	"	"	"

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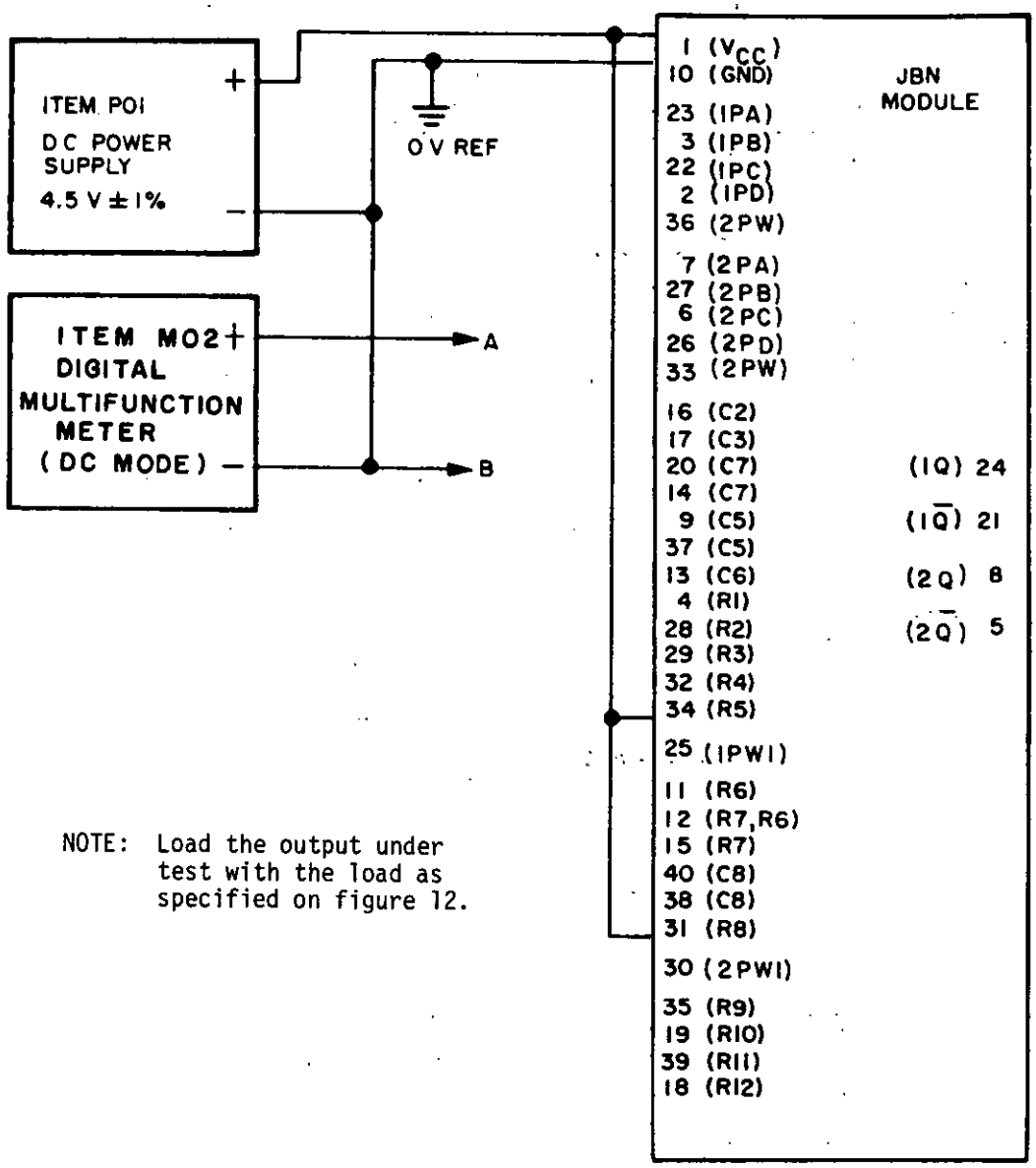
CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. Since items XOI-1 and XOI-2 are not used simultaneously, only one item XOI is required.
2. Since items POI-1, POI-2 and POI-3 are not used simultaneously, only two item POI are required.
3. With point "A" disconnected, item XOI-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

* FIGURE 10. Input current (high-level and low-level) and output short circuit current test circuit for table VII.

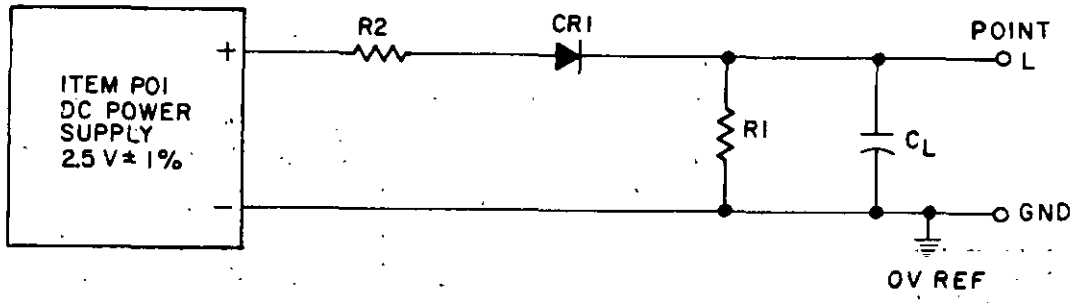
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NOTE: Load the output under test with the load as specified on figure 12.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for table VIII.

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NOTES:
1. R1 = 3.32 kΩ ±1%, 1/8 W.
2. R2 = Select value for a current flow of 9.6 mA ±5% from point L when L is held at 0.4 V.
3. CR1 = 1N4150, or equivalent.
4. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions H1-H4, J1-J4, Q1-Q102, R1-R4, S1-S4, T1-T16, U1-U16, and V1-V8.
5. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions X1-X4 and Y1-Y4.
6. C_L is not required for test conditions H1-H4 and J1-J4.

* FIGURE 12. JBN logic load circuit.

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TABLE IX. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K	Power supply current	15	50	mA	13

TABLE X. Resistance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	R1	Resistance	1	4	9,900	10,100	Ω	14
L2	R9	"	"	35	9,900	10,100	"	"
L3	R2	"	"	28	20,295	20,705	"	"
L4	R10	"	"	19	20,295	20,705	"	"
L5	R3	"	"	29	29,799	30,401	"	"
L6	R11	"	"	39	29,799	30,401	"	"
L7	R4	"	"	32	45,936	46,864	"	"
L8	R12	"	"	18	45,936	46,864	"	"
L9	R5	"	36	34	5,059	5,161	"	"
L10	R8	"	33	31	5,059	5,161	"	"
L11	R6	"	11	12	1,198	1,222	"	"
L12	R7	"	12	15	2,465	2,515	"	"

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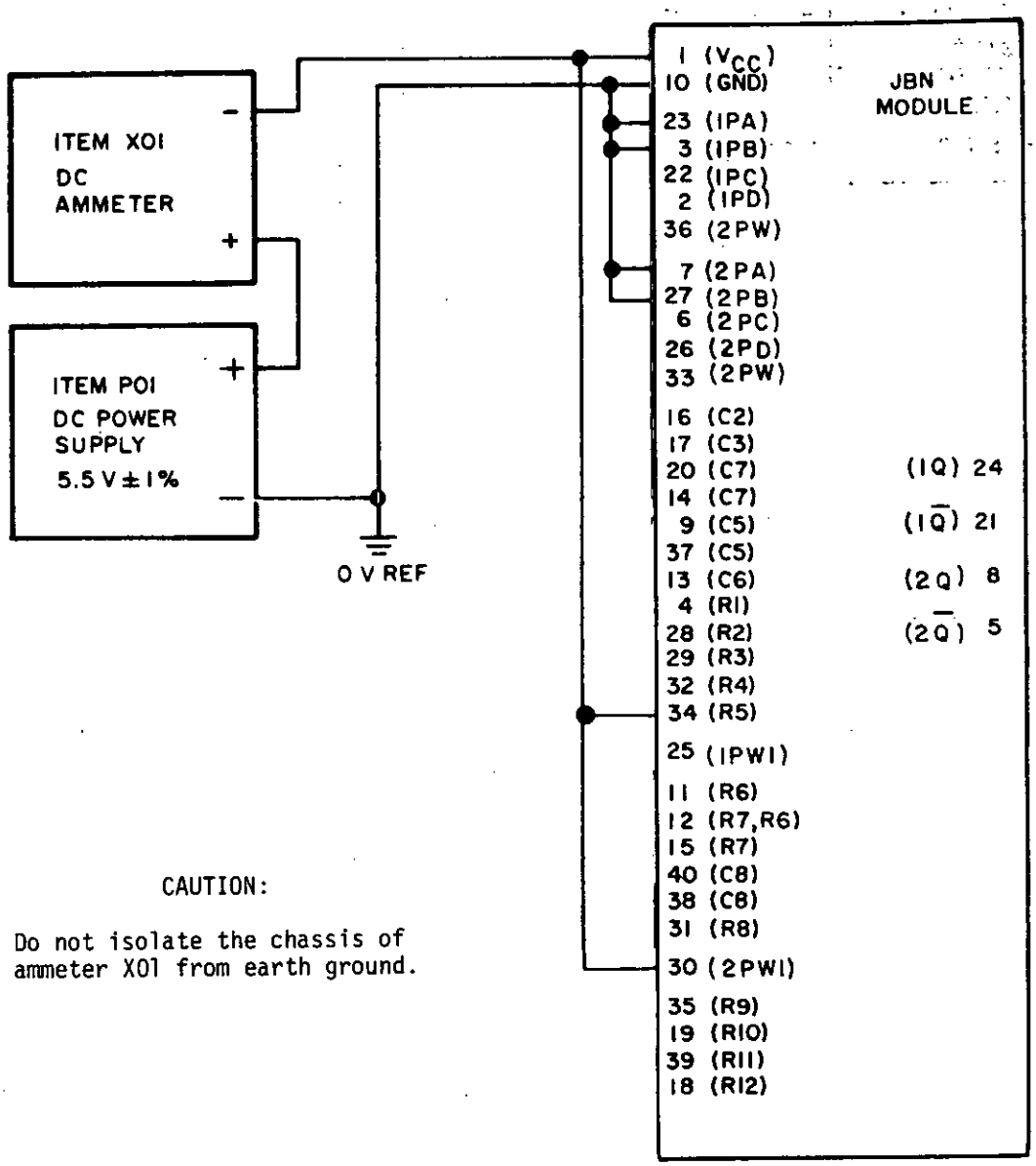


FIGURE 13. Power supply current test circuit for table IX.

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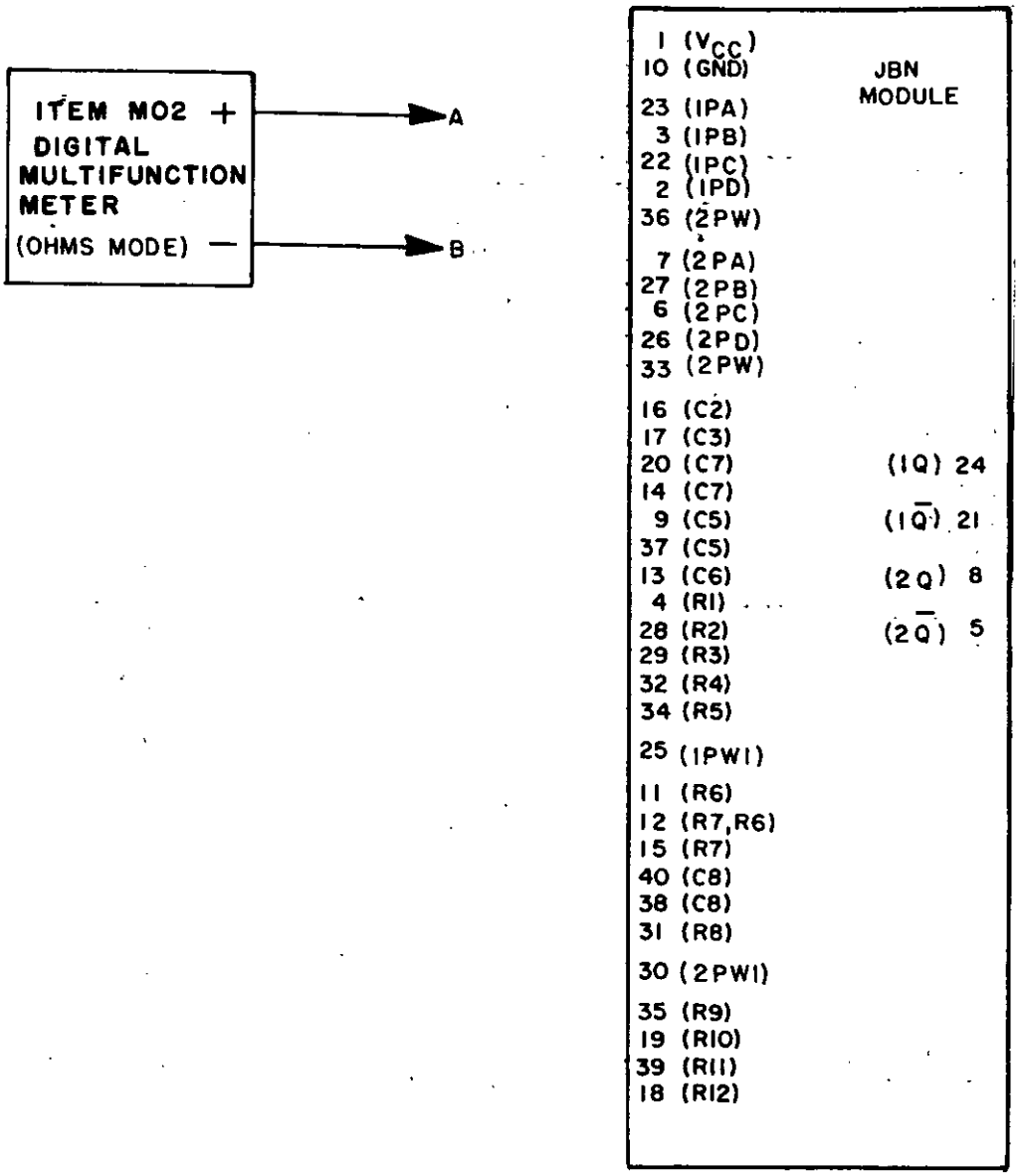


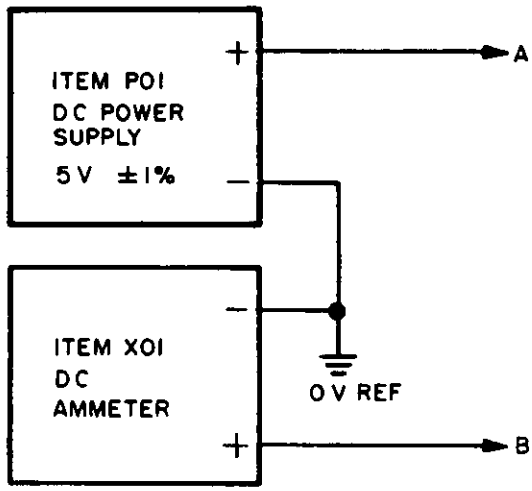
*FIGURE 14. Resistance test circuit for table X.

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TABLE XI. Capacitor dc leakage current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	C2 Capacitor dc leakage current	16	25	0	3	µA	15
M2	C3 "	17	25	"	"	"	"
M3	C5 "	9	37	"	"	"	"
M4	C6 "	13	30	"	"	"	"
M5	C7 "	20	14	"	"	"	"
M6	C8 "	40	38	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: The unit under test conforms to requirements if leakage drops below the limit in table XI in 5 minutes or less.

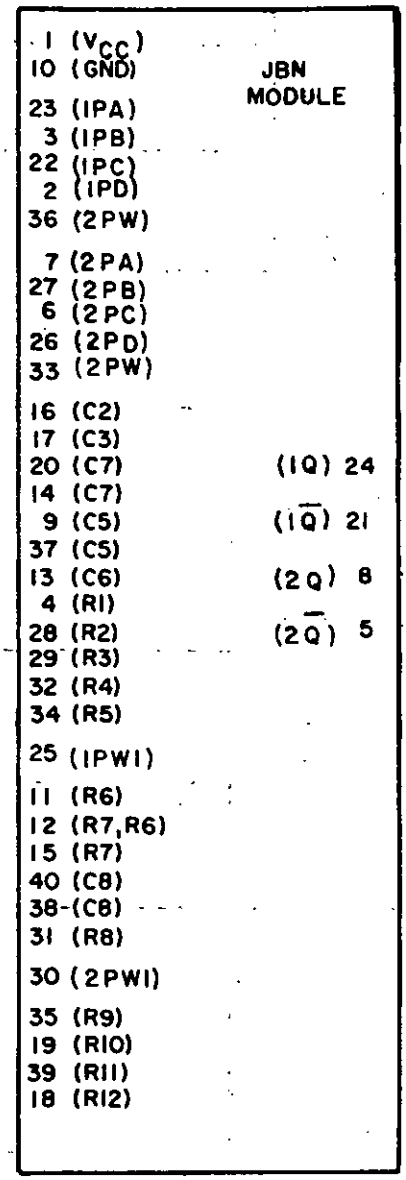


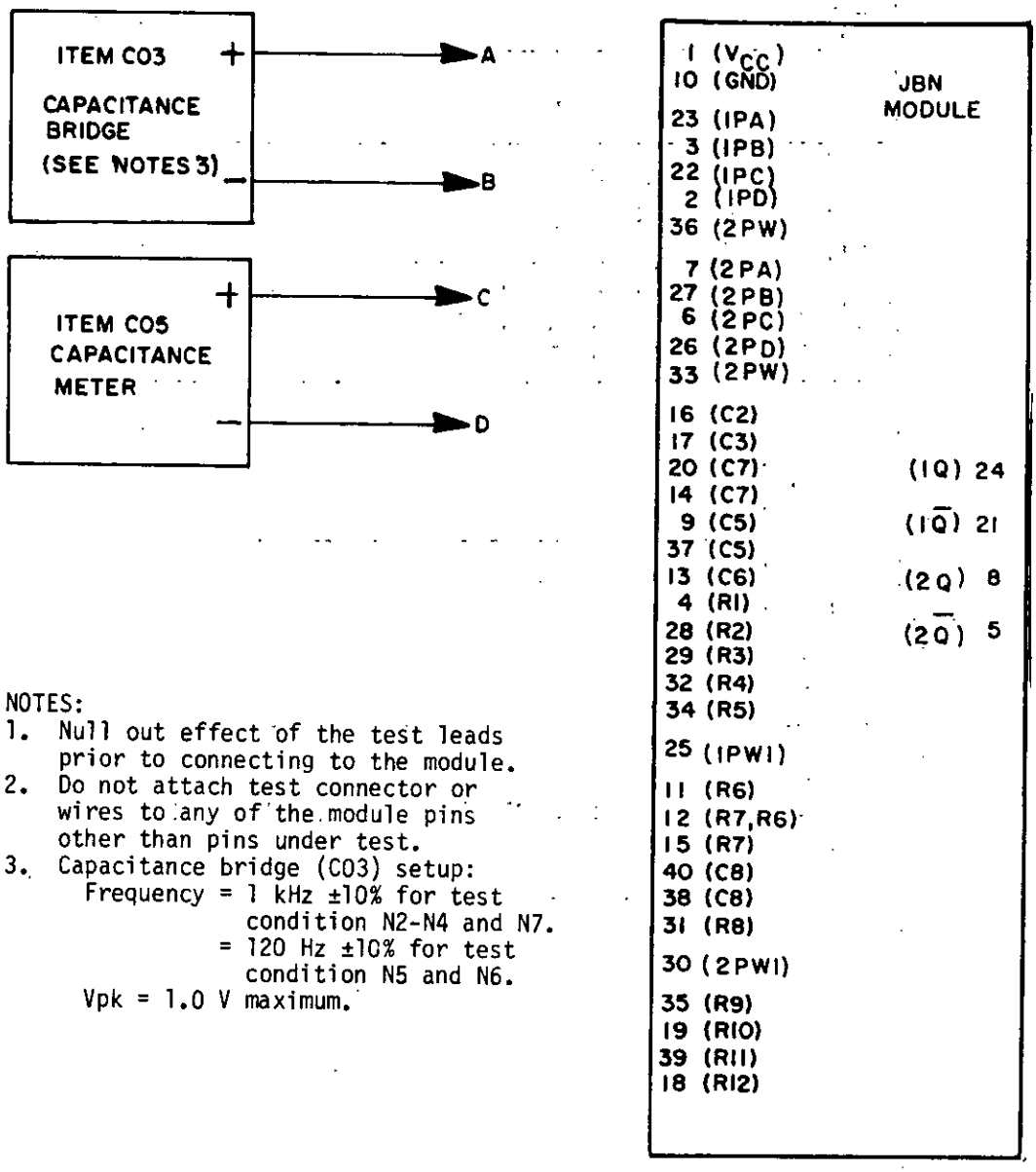
FIGURE 15. Capacitor dc leakage current test circuit for table XI.

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TABLE XII. Capacitance and capacitor dissipation factor test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	C1	Capacitance	Open	Open	36	25	90	130	pF	16
N2	C2	"	16	25	Open	Open	1,080	1,320	pF	"
N3	C3	"	17	25	"	"	0.009	0.011	µF	"
N4	C5	"	9	37	"	"	0.0738	0.0902	"	"
N5	C6	"	13	30	"	"	0.612	0.748	"	"
N6	C7	"	20	14	"	"	6.12	7.48	"	"
N7	C8	"	40	38	"	"	1,080	1,320	pF	"
P1	C2	Capacitor dissipation factor	16	25	"	"	0.0	2.5	%	"
P2	C3	"	17	25	"	"	"	"	"	"
P3	C5	"	9	37	"	"	"	"	"	"
P4	C8	"	40	38	"	"	"	"	"	"
P5	C6	"	13	30	"	"	0	3	"	"
P6	C7	"	20	14	"	"	0	3	"	"

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NOTES:
1. Null out effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pins under test.
3. Capacitance bridge (CO3) setup:
   Frequency = 1 kHz ±10% for test condition N2-N4 and N7.
   = 120 Hz ±10% for test condition N5 and N6.
   Vpk = 1.0 V maximum.

* FIGURE 16. Capacitance and capacitor dissipation factor test circuit for table XII.

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Check the source to verify that this is the current version before use.

TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q1	1Q Pulse width, Rx = 5.11 kΩ, Cx = 1,300 pF	2,3,22	23	24	Open	1 to 34, 16 to 36	1.86	2.98	µs	17,18
Q2	1$\overline{Q}$ "	2,3,22	23	21	"	"	"	"	"	"
Q3	1Q "	2,22,23	3	24	"	"	"	"	"	"
Q4	1$\overline{Q}$ "	2,22,23	3	21	"	"	"	"	"	"
Q5	1Q "	2,3	22	24	23	"	"	"	"	"
Q6	1$\overline{Q}$ "	2,3	22	21	23	"	"	"	"	"
Q7	1Q "	22,23	2	24	3	"	"	"	"	"
Q8	1$\overline{Q}$ "	22,23	2	21	3	"	"	"	"	"
Q9	2Q Pulse width, Rx = 5.11 kΩ, Cx = 1,200 pF	6,26,27	7	8	Open	1 to 31, 30 to 38, 33 to 40	1.72	2.74	"	"
Q10	2$\overline{Q}$ "	6,26,27	7	5	"	"	"	"	"	"
Q11	2Q "	6,7,26	27	8	"	"	"	"	"	"
Q12	2$\overline{Q}$ "	6,7,26	27	5	"	"	"	"	"	"
Q13	2Q "	7,26	6	8	27	"	"	"	"	"
Q14	2$\overline{Q}$ "	7,26	6	5	27	"	"	"	"	"
Q15	2Q "	6,27	26	8	7	"	"	"	"	"
Q16	2$\overline{Q}$ "	6,27	26	5	7	"	"	"	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q17	1Q Pulse width, Rx = 10 kΩ, Cx = 100 pF	2,3,22	23	24	Open	4 to 36	423	677	ns	17,18
Q18	1$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 100 pF	"	"	21	"	4 to 36	423	677	"	"
Q19	1Q Pulse width, Rx = 20.5 kΩ, Cx = 100 pF	"	"	24	"	28 to 36	847	1,353	"	"
Q20	1$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 100 pF	"	"	21	"	28 to 36	847	1,353	"	"
Q21	1Q Pulse width, Rx = 30.1 kΩ, Cx = 100 pF	"	"	24	"	29 to 36	1,155	1,845	"	"
Q22	1$\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 100 pF	"	"	21	"	29 to 36	1,155	1,845	"	"
Q23	1Q Pulse width, Rx = 46.4 kΩ, Cx = 100 pF	"	"	24	"	32 to 36	1,733	2,787	"	"
Q24	1$\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 100 pF	"	"	21	"	32 to 36	1,733	2,787	"	"
Q25	1Q Pulse width, Rx = 5.11 kΩ, Cx = 100 pF	"	"	24	"	1 to 34	223	357	"	"
Q26	1$\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = 100 pF	"	"	21	"	1 to 34	223	357	"	"
Q27	1Q Pulse width, Rx = 10 kΩ, Cx = 1,300 pF	"	"	24	"	4 to 36, 16 to 36	3.43	5.47	µs	"
Q28	1$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 1,300 pF	"	"	21	"	4 to 36, 16 to 36	3.43	5.47	µs	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q29	1Q Pulse width, Rx = 20.5 kΩ, Cx = 1,300 pF	2,3,22	23	24	Open	16 to 36, 28 to 36	6.79	10.85	µs	17,18
Q30	1$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 1,300 pF	"	"	21	"	16 to 36, 28 to 36	6.79	10.85	"	"
Q31	1Q Pulse width, Rx = 30.1 kΩ, Cx = 1,300 pF	"	"	24	"	16 to 36, 29 to 36	9.9	15.7	"	"
Q32	1$\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 1,300 pF	"	"	21	"	16 to 36, 29 to 36	9.9	15.7	"	"
Q33	1Q Pulse width, Rx = 46.4 kΩ, Cx = 1,300 pF	"	"	24	"	16 to 36, 32 to 36	15.1	24.1	"	"
Q34	1$\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 1,300 pF	"	"	21	"	16 to 36, 32 to 36	15.1	24.1	"	"
Q35	1Q Pulse width, Rx = 5.11 kΩ, Cx = 10,100 pF	"	"	24	"	1 to 34, 17 to 36	14.5	23.1	"	"
Q36	1$\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = 10,100 pF	"	"	21	"	1 to 34, 17 to 36	14.5	23.1	"	"
Q37	1Q Pulse width, Rx = 10 kΩ, Cx = 10,100 pF	"	"	24	"	4 to 36, 17 to 36	26.7	42.5	"	"
Q38	1$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 10,100 pF	"	"	21	"	4 to 36, 17 to 36	26.7	42.5	"	"
Q39	1Q Pulse width, Rx = 20.5 kΩ, Cx = 10,100 pF	"	"	24	"	17 to 36, 28 to 36	52.8	84.2	"	"
Q40	1$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 10,100 pF	"	"	21	"	17 to 36, 28 to 36	52.8	84.2	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q41	1Q Pulse width, Rx = 30.1 kΩ, Cx = 10,100 pF	2,3,22	23	24	Open	17 to 36, 29 to 36	76.7	112.5	µs	17,18
Q42	1$\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 10,100 pF	"	"	21	"	17 to 36, 29 to 36	76.7	112.5	"	"
Q43	1Q Pulse width, Rx = 46.4 kΩ, Cx = 10,100 pF	"	"	24	"	17 to 36, 32 to 36	117.2	187.2	"	"
Q44	1$\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 10,100 pF	"	"	21	"	17 to 36, 32 to 36	117.2	187.2	"	"
Q45	1Q Pulse width, Rx = 5.11 kΩ, Cx = 6.8 µF	"	"	24	"	1 to 34, 14 to 25, 20 to 36	9.7	15.5	ms	"
Q46	1$\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = 6.8 µF	"	"	21	"	1 to 34, 14 to 25, 20 to 36	9.7	15.5	"	"
Q47	1Q Pulse width, Rx = 10 kΩ, Cx = 6.8 µF	"	"	24	"	4 to 36, 14 to 25, 20 to 36	17.9	28.7	"	"
Q48	1$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 6.8 pF	"	"	21	"	4 to 36, 14 to 25, 20 to 36	17.9	28.7	"	"
Q49	1Q Pulse width, Rx = 20.5 kΩ, Cx = 6.8 µF	"	"	24	"	14 to 25, 20 to 36, 28 to 36	35.5	56.7	"	"
Q50	1$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 6.8 µF	"	"	21	"	14 to 25, 20 to 36, 28 to 36	35.5	56.7	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q51	1Q Pulse width, Rx = 30.1 kΩ, Cx = 6.8 µF	2,3,22	23	24	Open	14 to 25, 20 to 36, 29 to 36	51.6	82.4	ms	17,18
Q52	1$\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 6.8 pF	"	"	21	"	14 to 25, 20 to 36, 29 to 36	51.6	82.4	"	"
Q53	1Q Pulse width, Rx = 46.4 kΩ, Cx = 6.8 µF	"	"	24	"	14 to 25, 20 to 36, 32 to 36	79	126	"	"
Q54	1$\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 6.8 µF	"	"	21	"	14 to 25, 20 to 36, 32 to 36	79	126	"	"
Q55	2Q Pulse width, Rx = 5.11 kΩ, Cx = Open	6,26,27	7	8	"	1 to 31	46	101	ns	"
Q56	2$\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = Open	"	"	5	"	1 to 31	46	101	"	"
Q57	2Q Pulse width, Rx = 10 kΩ, Cx = Open	"	"	8	"	33 to 35	116	185	"	"
Q58	2$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = Open	"	"	5	"	33 to 35	116	185	"	"
Q59	2Q Pulse width, Rx = 20.5 kΩ, Cx = Open	"	"	8	"	19 to 33	216	344	"	"
Q60	2$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = Open	"	"	5	"	19 to 33	216	344	"	"
Q61	2Q Pulse width, Rx = 30.1 kΩ, Cx = Open	"	"	8	"	33 to 39	308	492	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q62	$2\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = Open	6,26,27	7	5	Open	33 to 39	308	492	ns	17,18
Q63	2Q Pulse width, Rx = 46.4 kΩ, Cx = Open	"	"	8	"	18 to 33	477	763	"	"
Q64	$2\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = Open	"	"	5	"	18 to 33	477	763	"	"
Q65	2Q Pulse width, Rx = 10 kΩ, Cx = 1,200 pF	"	"	8	"	30 to 38, 33 to 35, 33 to 40	3.17	5.07	µs	"
Q66	$2\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 1,200 pF	"	"	5	"	30 to 38, 33 to 35, 33 to 40	3.17	5.07	"	"
Q67	2Q Pulse width, Rx = 20.5 kΩ, Cx = 1,200 pF	"	"	8	"	19 to 33, 30 to 38, 33 to 40	6.27	10.01	"	"
Q68	$2\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 1,200 pF	"	"	5	"	19 to 33, 30 to 38, 33 to 40	6.27	10.01	"	"
Q69	2Q Pulse width, Rx = 30.1 kΩ, Cx = 1,200 pF	"	"	8	"	30 to 38, 33 to 39, 33 to 40	9.1	14.5	"	"
Q70	$2\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 1,200 pF	"	"	5	"	30 to 38, 33 to 39, 33 to 40	9.1	14.5	"	"
Q71	2Q Pulse width, Rx = 46.4 kΩ, Cx = 1,200 pF	"	"	8	"	18 to 33, 30 to 38, 33 to 40	13.9	22.3	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q72	$2\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 1,200 pF	6,26,27	7	5	Open	18 to 33, 30 to 38, 33 to 40	13.9	22.3	µs	17,18
Q73	2Q Pulse width, Rx = 5.11 kΩ, Cx = 82,000 pF	"	"	8	"	1 to 31, 9 to 33, 30 to 37	117	187	"	"
Q74	$2\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = 82,000 pF	"	"	5	"	1 to 31, 9 to 33, 30 to 37	117	187	"	"
Q75	2Q Pulse width, Rx = 10 kΩ, Cx = 82,000 pF	"	"	8	"	9 to 33, 30 to 37, 33 to 35	216	346	"	"
Q76	$2\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 82,000 pF	"	"	5	"	9 to 33, 30 to 37, 33 to 35	216	346	"	"
Q77	2Q Pulse width, Rx = 20.5 kΩ, Cx = 82,000 pF	"	"	8	"	9 to 33, 19 to 33, 30 to 37	428	684	"	"
Q78	$2\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 82,000 pF	"	"	5	"	9 to 33, 19 to 33, 30 to 37	428	684	"	"
Q79	2Q Pulse width, Rx = 30.1 kΩ, Cx = 82,000 pF	"	"	8	"	9 to 33, 30 to 37, 33 to 39	622	994	"	"
Q80	$2\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 82,000 pF	"	"	5	"	9 to 33, 30 to 37, 33 to 39	622	994	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q81	2Q Pulse width, Rx = 46.4 kΩ, Cx = 82,000 pF	6,26,27	7	8	Open	9 to 33, 18 to 33, 30 to 37	952	1,520	µs	17,18
Q82	2$\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 82,000 pF	"	"	5	"	9 to 33, 18 to 33, 30 to 37	952	1,520	µs	"
Q83	2Q Pulse width, Rx = 5.11 kΩ, Cx = 0.68 µF	"	"	8	"	1 to 31, 13 to 33	0.97	1.55	ms	"
Q84	2$\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = 0.68 µF	"	"	5	"	1 to 31, 13 to 33	0.97	1.55	"	"
Q85	2Q Pulse width, Rx = 10 kΩ, Cx = 0.68 µF	"	"	8	"	13 to 33, 35 to 33	1.79	2.87	"	"
Q86	2$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 0.68 µF	"	"	5	"	13 to 33, 35 to 33	1.79	2.87	"	"
Q87	2Q Pulse width, Rx = 20.5 kΩ, Cx = 0.68 µF	"	"	8	"	13 to 33, 19 to 33	3.55	5.67	"	"
Q88	2$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 0.68 µF	"	"	5	"	13 to 33, 19 to 33	3.55	5.67	"	"
Q89	2Q Pulse width, Rx = 30.1 kΩ, Cx = 0.68 µF	"	"	8	"	13 to 33, 33 to 39	5.16	8.24	"	"
Q90	2$\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 0.68 µF	"	"	5	"	13 to 33, 33 to 39	5.16	8.24	"	"
Q91	2Q Pulse width, Rx = 46.4 kΩ, Cx = 0.68 µF	"	"	8	"	13 to 33, 18 to 33	7.89	12.61	"	"
Q92	2$\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 0.68 µF	"	"	5	"	13 to 33, 18 to 33	7.89	12.61	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q93	2Q Pulse width, Rx = 5.11 kΩ, Cx = 100 pF	6,26,27	7	8	Open	1 to 31, E to 33, F to 30	223	357	ns	17,18
Q94	2$\overline{Q}$ Pulse width, Rx = 5.11 kΩ, Cx = 100 pF	"	"	5	"	1 to 31, E to 33, F to 30	223	357	"	"
Q95	2Q Pulse width, Rx = 10 kΩ, Cx = 100 pF	"	"	8	"	33 to 35, E to 33, F to 30	423	677	"	"
Q96	2$\overline{Q}$ Pulse width, Rx = 10 kΩ, Cx = 100 pF	"	"	5	"	33 to 35, E to 33, F to 30	423	677	"	"
Q97	2Q Pulse width, Rx = 10 kΩ, Cx = 100 pF	"	"	8	"	19 to 33, E to 33 F to 30	847	1,353	"	"
Q98	2$\overline{Q}$ Pulse width, Rx = 20.5 kΩ, Cx = 100 pF	"	"	5	"	19 to 33, E to 33 F to 30	847	1,353	"	"
Q99	2Q Pulse width, Rx = 30.1 kΩ, Cx = 100 pF	"	"	8	"	33 to 39, E to 33, F to 30	1,155	1,845	"	"
Q100	2$\overline{Q}$ Pulse width, Rx = 30.1 kΩ, Cx = 100 pF	"	"	5	"	33 to 39, E to 33, F to 30	1,155	1,845	"	"
Q101	2Q Pulse width, Rx = 46.4 kΩ, Cx = 100 pF	"	"	8	"	18 to 33, E to 33 F to 30	1,733	2,787	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
Q102	$2\overline{Q}$ Pulse width, Rx = 46.4 kΩ, Cx = 100 pF	6,26,27	7	5	Open	18 to 33, E to 33, F to 30	1,733	2,787	ns	17,18
R1	1Q Transition time (t_{TLH})	2,3,22	23	24	"	1 to 34, 16 to 36	10	80	"	17,19
R2	$1\overline{Q}$ "	2,3,22	23	21	"	1 to 34, 16 to 36	"	"	"	"
R3	2Q "	6,26,27	7	8	"	1 to 31, 30 to 38, 33 to 40	"	"	"	"
R4	$2\overline{Q}$ "	6,26,27	7	5	"	1 to 31, 30 to 38, 33 to 40	"	"	"	"
S1	1Q Transition time (t_{THL})	2,3,22	23	24	"	1 to 34, 16 to 36	2	25	"	"
S2	$1\overline{Q}$ "	2,3,22	23	21	"	1 to 34, 16 to 36	"	"	"	"
S3	2Q "	6,26,27	7	8	"	1 to 31, 30 to 38, 33 to 40	"	"	"	"
S4	$2\overline{Q}$ "	6,26,27	7	5	"	1 to 31, 30 to 38, 33 to 40	"	"	"	"
T1	1Q Propagation delay time (t_{PLH}), minimum setup and hold time	2,3,22	23	24	"	1 to 34, 16 to 36	5	49	"	17,18

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

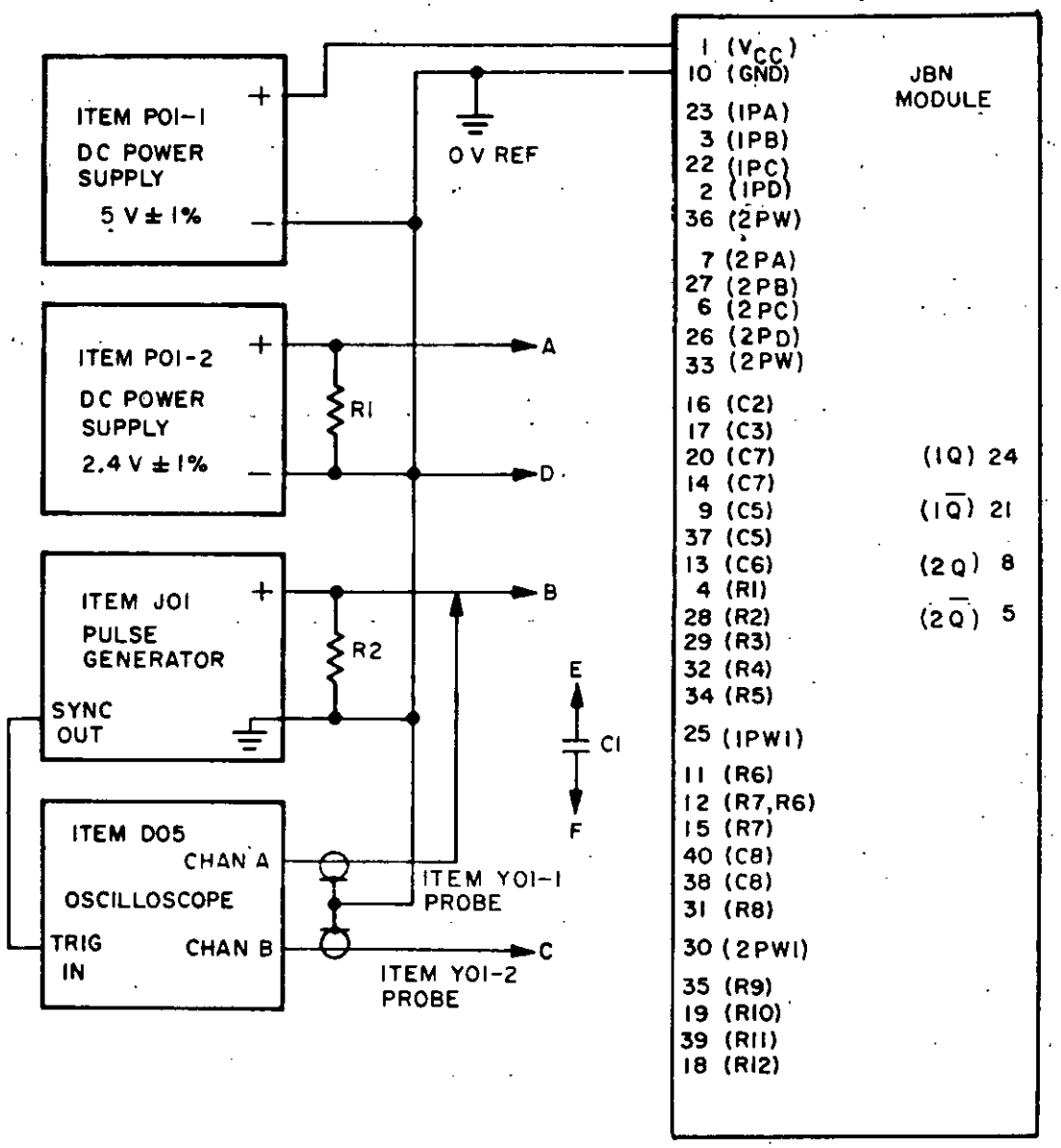
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
T2	1Q Propagation delay time (t_{PLH}), minimum setup and hold time	2,22,23	3	24	Open	1 to 34, 16 to 36	5	49	ns	17,18
T3	1Q "	2,3	22	"	23	"	"	"	"	17,18
T4	1Q "	22,23	2	"	3	"	"	"	"	17,18
T5	1Q "	2,3,22	23	"	Open	"	"	"	"	17,19
T6	1Q "	2,22,23	3	"	Open	"	"	"	"	"
T7	1Q "	2,3	22	"	23	"	"	"	"	"
T8	1Q "	22,23	2	"	3	"	"	"	"	"
T9	2Q "	6,26,27	7	8	Open	1 to 31, 30 to 38, 33 to 40	"	"	"	17,18
T10	2Q "	6,7,26	27	"	Open	"	"	"	"	"
T11	2Q "	7,26	6	"	27	"	"	"	"	"
T12	2Q "	6,27	26	"	7	"	"	"	"	"
T13	2Q "	6,26,27	7	"	Open	"	"	"	"	17,19
T14	2Q "	6,7,26	27	"	Open	"	"	"	"	"
T15	2Q "	7,26	6	"	27	"	"	"	"	"
T16	2Q "	6,27	26	"	7	"	"	"	"	"

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TABLE XIII. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
U1	$1\overline{Q}$ Propagation delay time (t_{PHL}), minimum setup and hold time	2,3,22	23	21	Open	1 to 34, 16 to 36	5	49	ns	17,18
U2	$1\overline{Q}$ "	2,22,23	3	"	Open	"	"	"	"	"
U3	$1\overline{Q}$ "	2,3	22	"	23	"	"	"	"	"
U4	$1\overline{Q}$ "	22,23	2	"	3	"	"	"	"	"
U5	$1\overline{Q}$ "	2,3,22	23	"	Open	"	"	"	"	17,19
U6	$1\overline{Q}$ "	2,22,23	3	"	Open	"	"	"	"	"
U7	$1\overline{Q}$ "	2,3	22	"	23	"	"	"	"	"
U8	$1\overline{Q}$ "	22,23	2	"	3	"	"	"	"	"
U9	$2\overline{Q}$ "	6,26,27	7	5	Open	1 to 31, 30 to 38, 33 to 40	"	"	"	17,18
U10	$2\overline{Q}$ "	6,7,26	27	"	Open	"	"	"	"	"
U11	$2\overline{Q}$ "	7,26	6	"	27	"	"	"	"	"
U12	$2\overline{Q}$ "	6,27	26	"	7	"	"	"	"	"
U13	$2\overline{Q}$ "	6,26,27	7	"	Open	"	"	"	"	17,19
U14	$2\overline{Q}$ "	6,7,26	27	"	Open	"	"	"	"	"
U15	$2\overline{Q}$ "	7,26	6	"	27	"	"	"	"	"
U16	$2\overline{Q}$ "	6,27	26	"	7	"	"	"	"	"

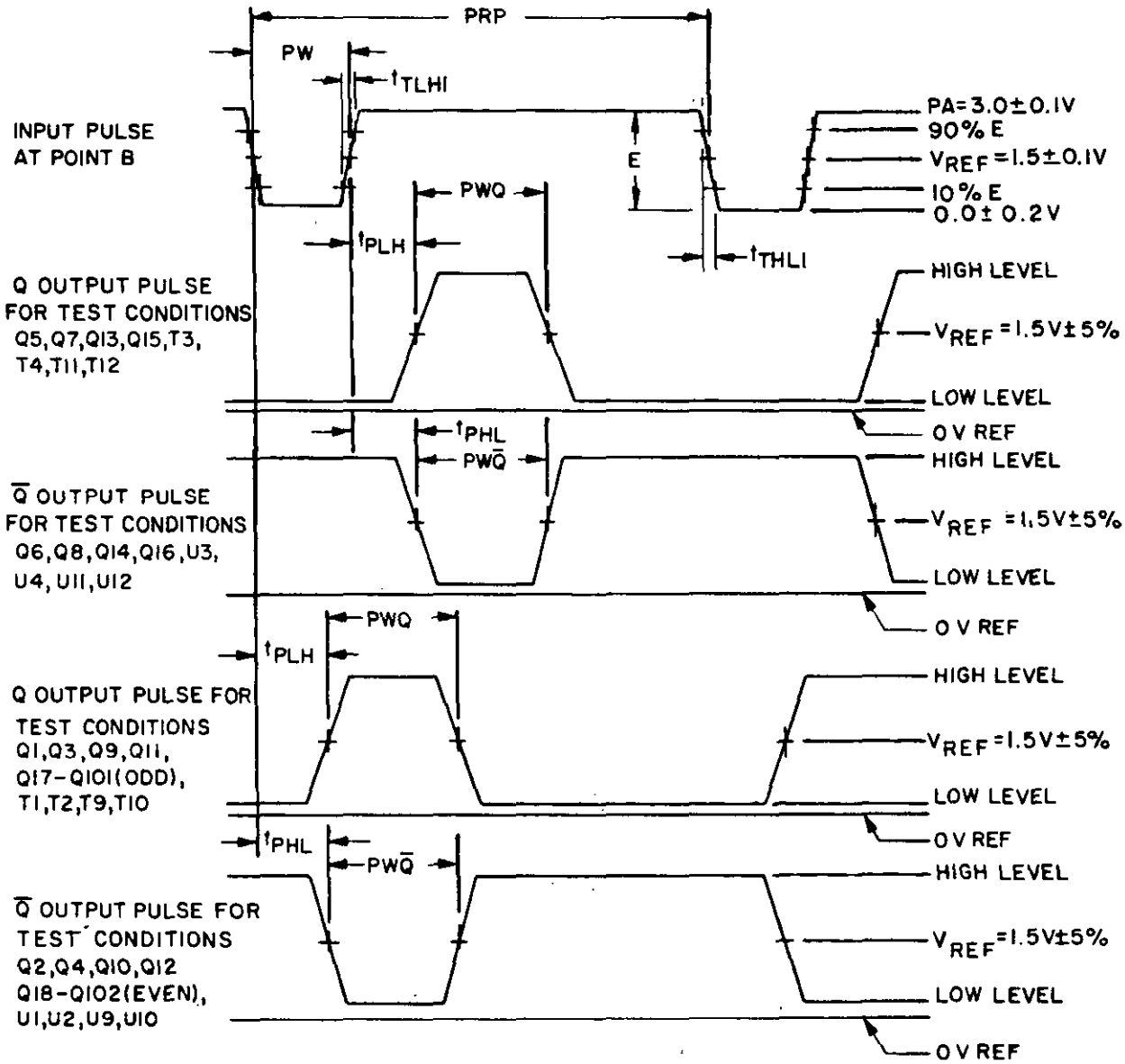
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. C1 = 100 pF ±10%, 100 V.
3. Load all outputs with the load as specified on figure 12.

FIGURE 17. Pulse width, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIII.

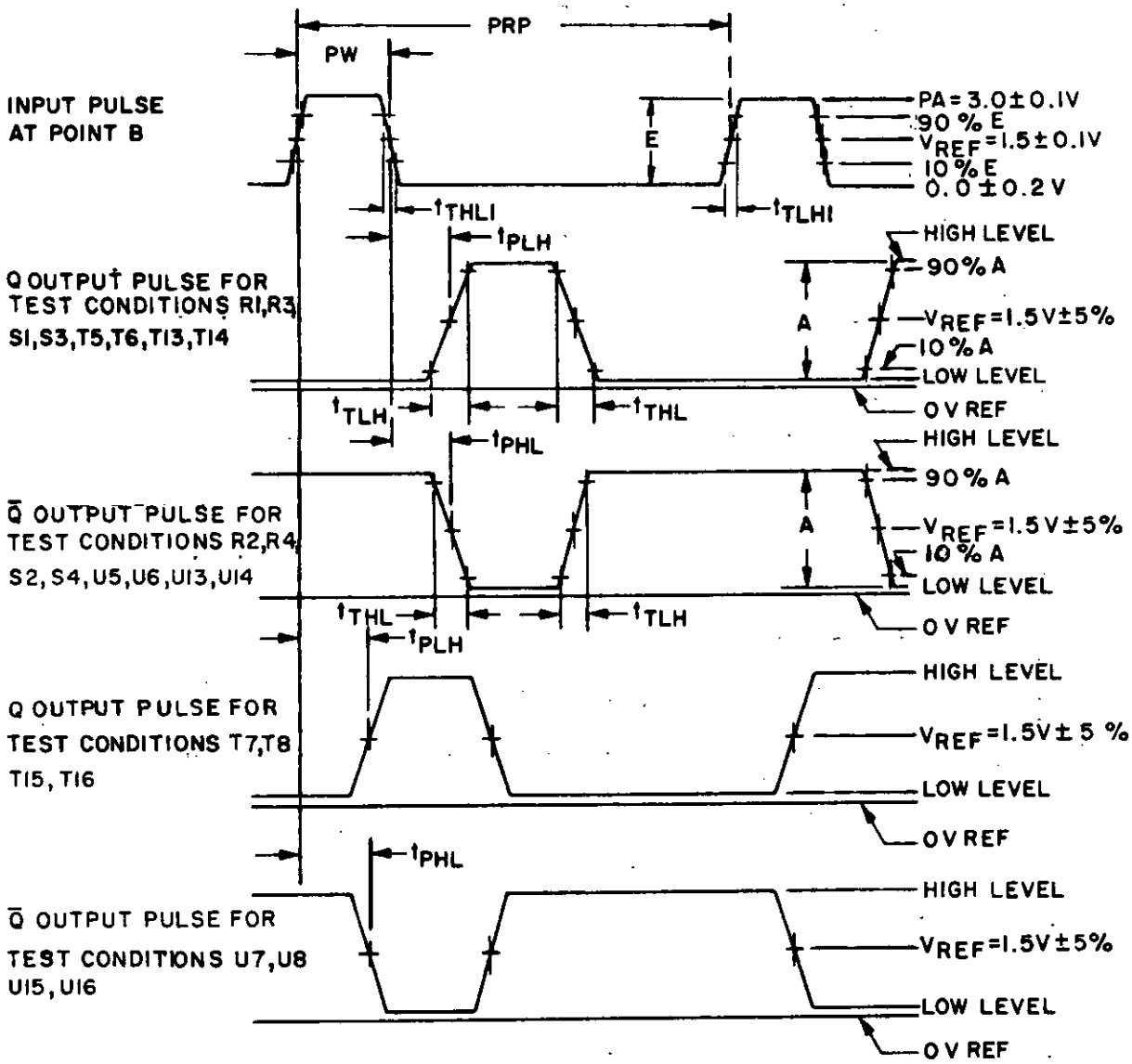
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NOTE: Input pulse parameters:
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
PW = 40 +2, -0 ns for test conditions Q1-Q16, T1-T4, T9-T12, U1-U4, and U9-U12.
= 45 ±4 ns for test conditions Q55-Q60.
= 200 ±20 ns for test conditions Q17-Q26, and Q93-Q102.
= 3.0 ±0.3 µs for test conditions Q27-Q38, and Q65-Q72.
= 50 ±5 µs for test conditions Q39-Q44, and Q73-Q80.
= 900 ±90 µs for test conditions Q81-Q92.
= 10 ±1 ms for test conditions Q45-Q54.
= 45 ±4 ns for test conditions Q61-Q64.
PRP = 3.5 ±0.3 µs for test conditions Q1-Q26, Q93-Q102, T1-T4, T9-T12, U1-U4, and U9-U12.
= 500 ±50 ns for test conditions Q55-Q60.
= 55 ±5 µs for test conditions Q27-Q38, Q65-Q72.
= 1.2 ±0.1 ms for test conditions Q39-Q44, and Q73-Q80.
= 15 ±1 ms for test conditions Q81-Q92.
= 150 ±10 ms for test conditions Q45-Q54.
= 1.0 ±0.1 µs for test conditions Q61-Q64.

*FIGURE 18. Pulse width and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIII.

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NOTE: Input pulse parameters:
$t_{TLHI} = t_{THLI} = 10 \pm 2$ ns.
PW = $40\ ^{+2}_{-0}$ ns.
PRP = 3.5 ±0.3 µs.

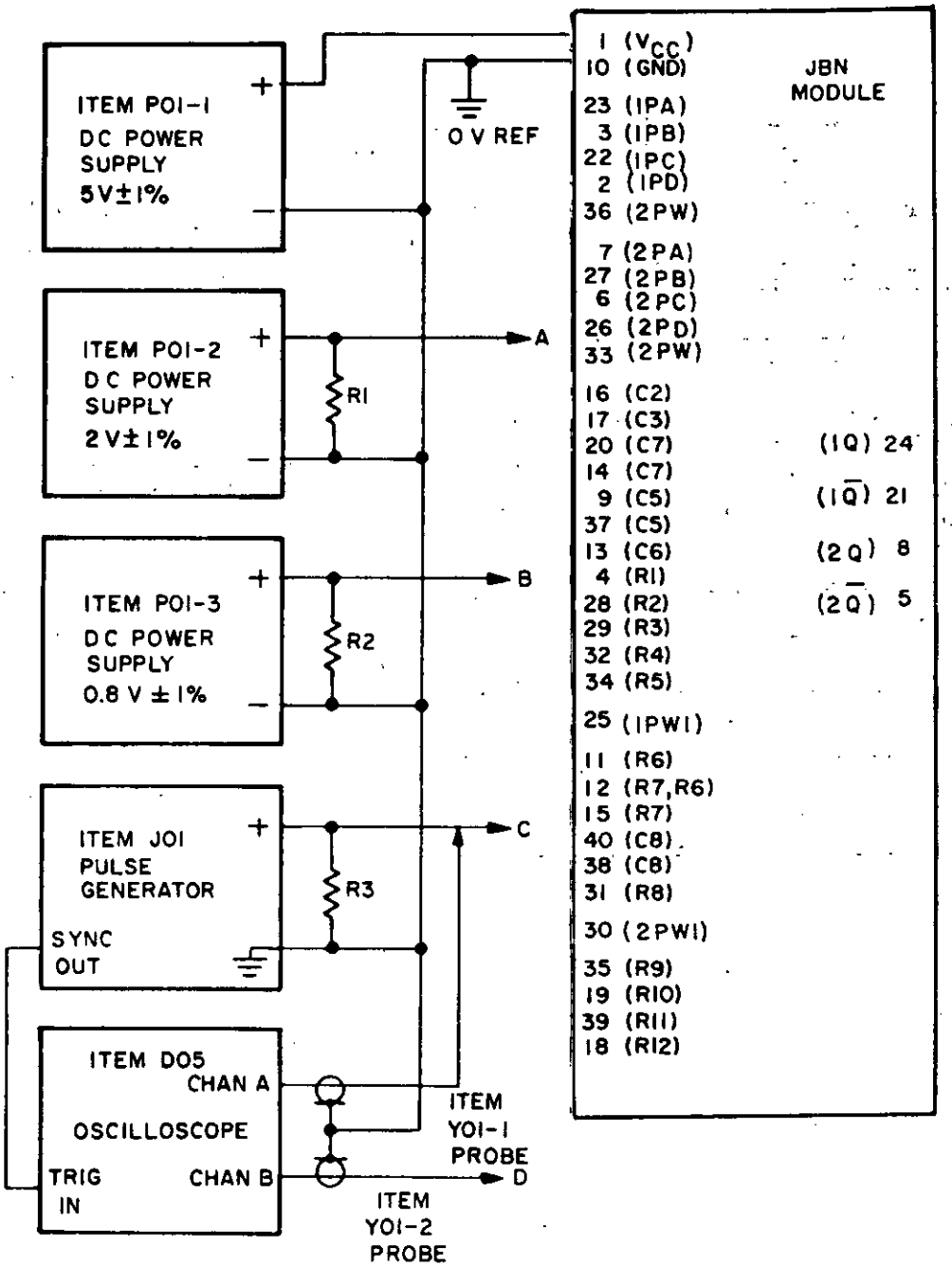
* FIGURE 19. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIII.

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TABLE XIV. Retrigger pulse width test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
V1	1Q Retrigger pulse width, Rx = 5.11 kΩ, Cx = 100 pF	2,3,22	Open	23	24	1 to 34	228	406	ns	20,21
V2	1$\overline{Q}$ "	2,3,22	"	23	21	"	"	"	"	"
V3	1Q "	2,22,23	"	3	24	"	"	"	"	"
V4	1$\overline{Q}$ "	2,22,23	"	3	21	"	"	"	"	"
V5	1Q "	2,3	23	22	24	"	"	"	"	"
V6	1$\overline{Q}$ "	2,23	3	22	21	"	"	"	"	"
V7	1Q "	3,22	23	2	24	"	"	"	"	"
V8	1$\overline{Q}$ "	22,23	3	2	21	"	"	"	"	"
V9	2Q Retrigger pulse width, Rx = 5.11 kΩ, Cx = 1,200 pF	6,26,27	Open	7	8	1 to 31, 30 to 38, 33 to 40	1.72	2.74	µs	"
V10	2$\overline{Q}$ "	6,26,27	"	7	5	"	"	"	"	"
V11	2Q "	6,7,26	"	27	8	"	"	"	"	"
V12	2$\overline{Q}$ "	6,7,26	"	27	5	"	"	"	"	"
V13	2Q "	26,27	7	6	8	"	"	"	"	"
V14	2$\overline{Q}$ "	7,26	27	6	5	"	"	"	"	"
V15	2Q "	6,27	7	26	8	"	"	"	"	"
V16	2$\overline{Q}$ "	6,7	27	26	5	"	"	"	"	"

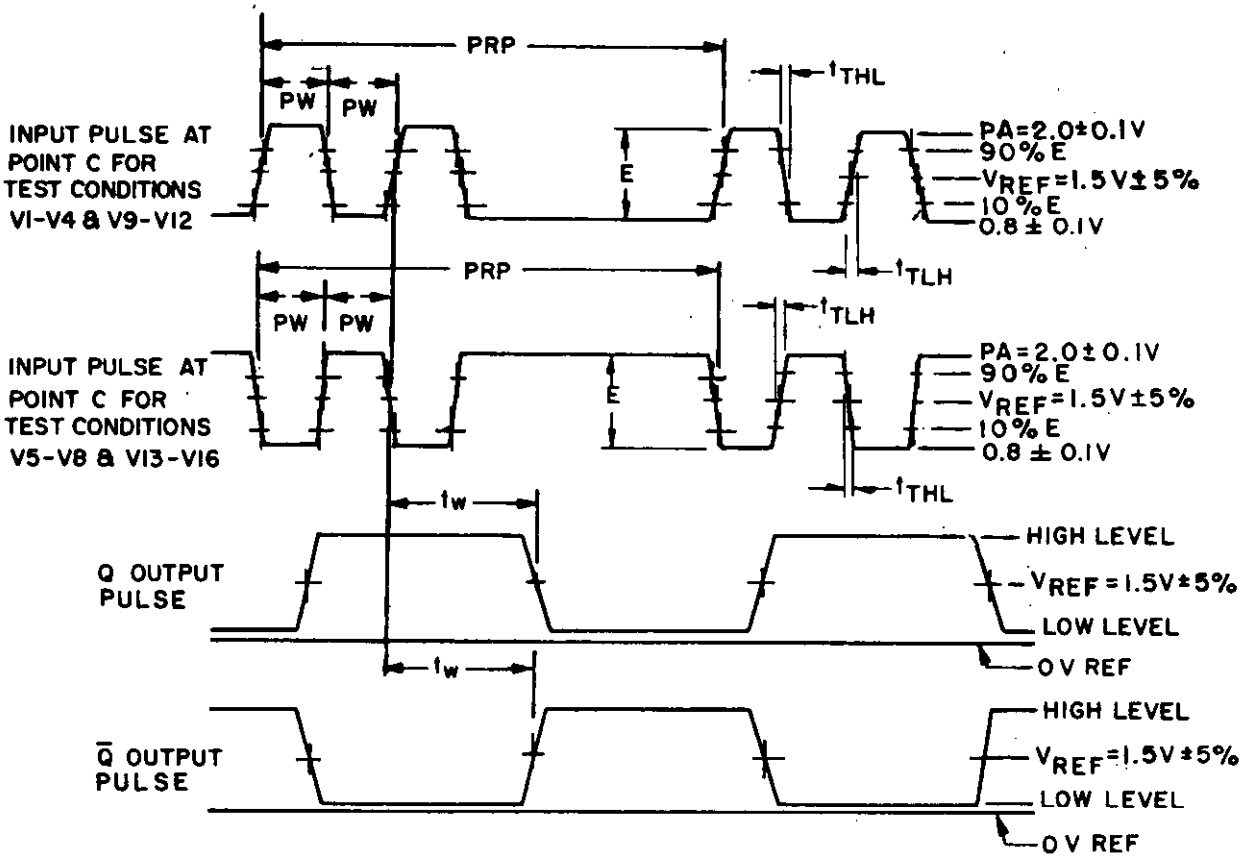
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs with the load as specified on figure 12.

FIGURE 20. Retrigger pulse width test circuit for table XIV.

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NOTES:

1. Input pulse parameters for test conditions V1-V8:
   a. PW = 40 $^{+4}_{-0}$ ns.
   b. PRP = 600 ±60 ns.
   c. $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Input pulse pulse parameters for test conditions V9-V16:
   a. PW = 400 ±40 ns.
   b. PRP = 4.0 ±0.4 µs.
   c. $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. Place item J01 in double pulse mode.

* FIGURE 21. Retrigger pulse width test circuit waveforms for table XIV.

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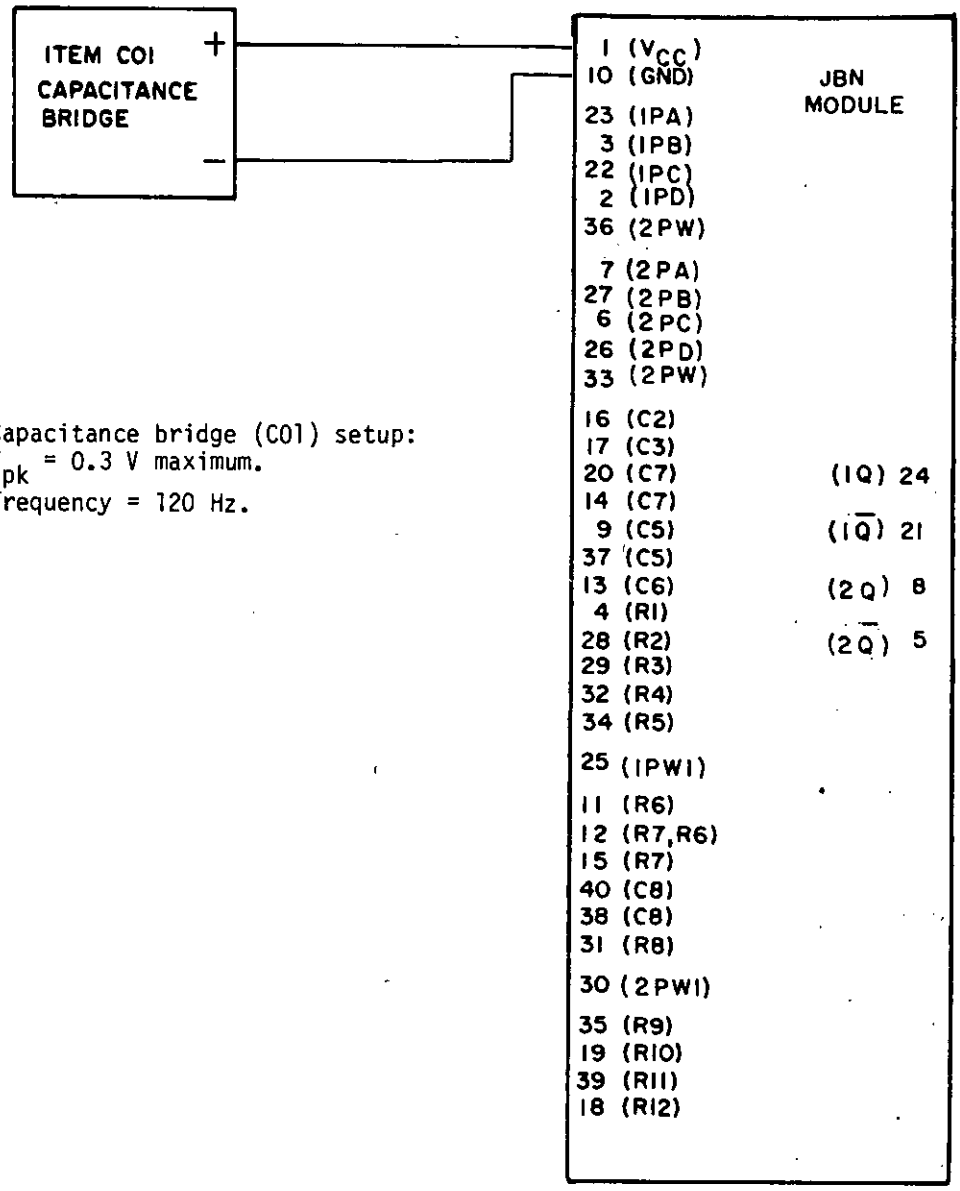
TABLE XV. Power supply filter capacitance test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
W	C4	Power supply capacitance	1.6	3.0	µF	22

TABLE XVI. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Pin to pin	Min	Max		
X1	1Q	Output crosstalk, high level	7,25	6,27	26	24	1 to 34, 33 to 39	0	400	mV	23,24
X2	$1\overline{Q}$	"	7	6,27	26	21	1 to 34, 33 to 39	"	"	"	"
X3	2Q	"	23,30	2,3	22	8	1 to 31, 1 to 34	"	"	"	"
X4	$2\overline{Q}$	"	23	2,3	22	5	1 to 31, 1 to 34	"	"	"	"
Y1	1Q	Output crosstalk, low level	7	6,27	26	24	1 to 34, 33 to 39	"	"	"	"
Y2	$1\overline{Q}$	"	7,25	6,27	26	21	1 to 34, 33 to 39	"	"	"	"
Y3	2Q	"	23	2,3	22	8	1 to 31, 1 to 34	"	"	"	"
Y4	$2\overline{Q}$	"	23,30	2,3	22	5	1 to 31, 1 to 34	"	"	"	"

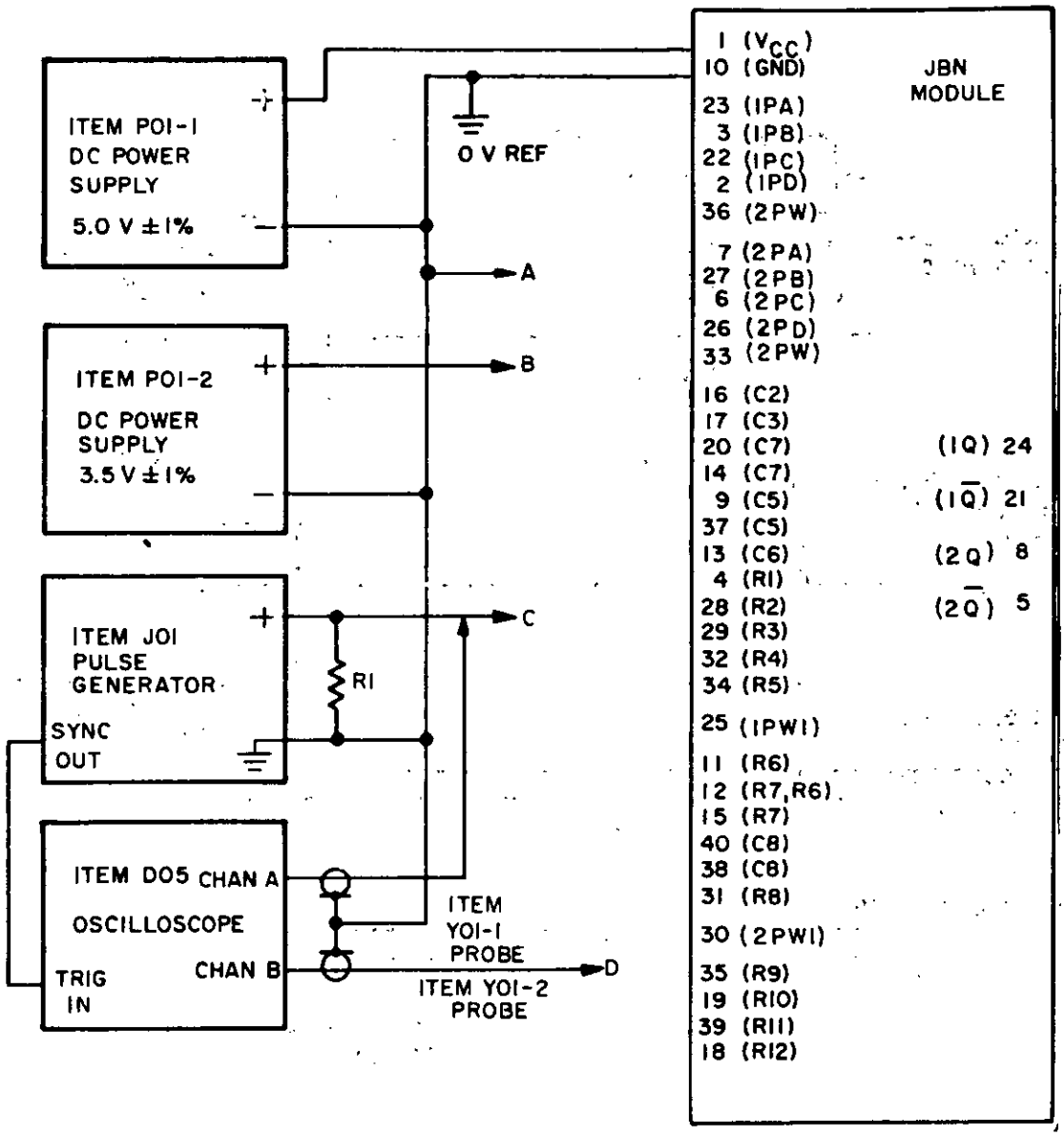
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NOTE: Capacitance bridge (C01) setup:
V_{pk} = 0.3 V maximum.
Frequency = 120 Hz.

* FIGURE 22. Power supply filter capacitance test circuit for table XV.

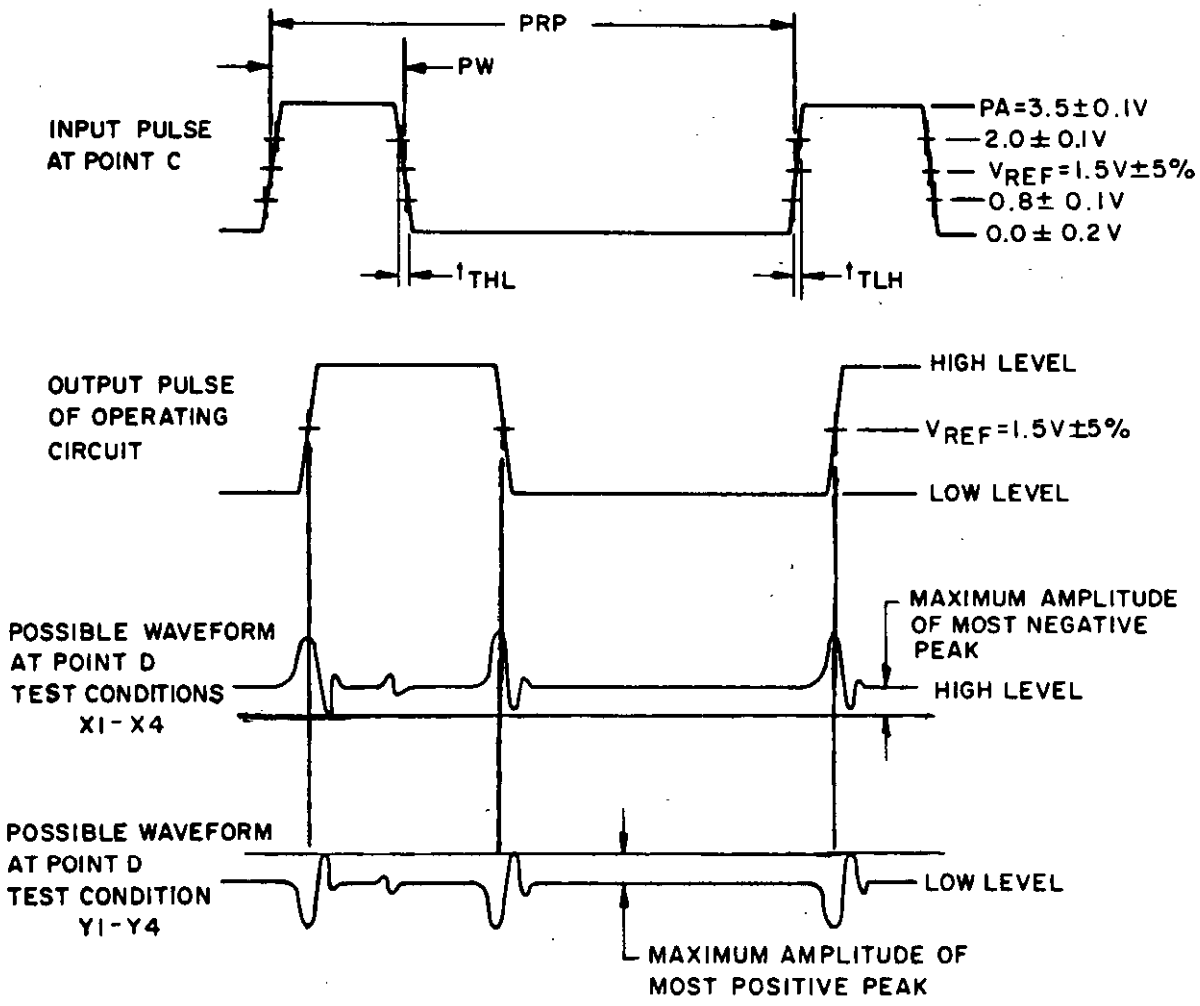
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NOTES:
1. RI = 51Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 12.

FIGURE 23. Output crosstalk (high-level and low-level) test circuit for table XVI.

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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
PRP = 800 ±80 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

* FIGURE 24. Output crosstalk (high-level and low-level) test circuit waveforms for table XVI.

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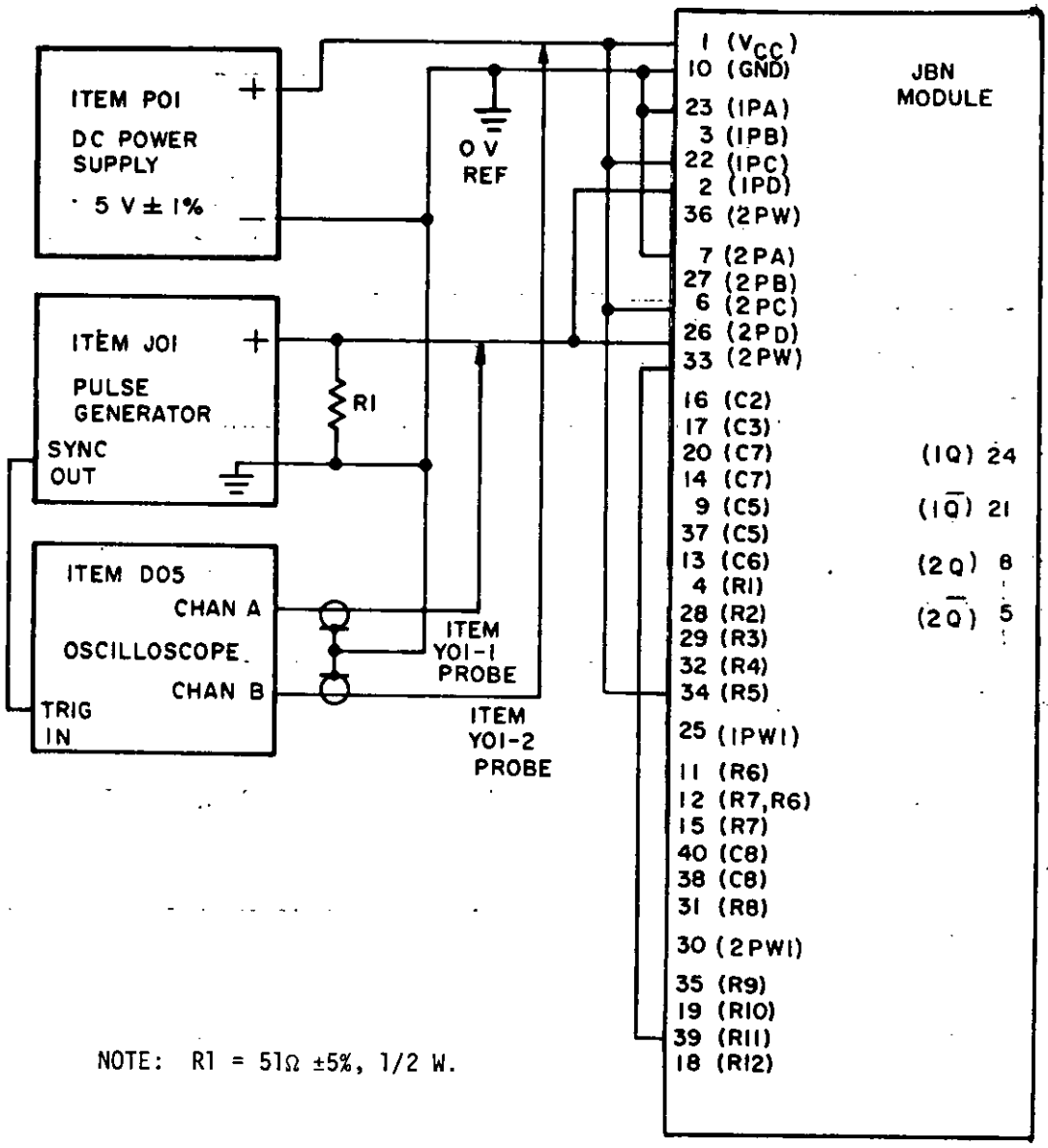
TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient amplitude	0	400	mV	25,26

TABLE XVIII. Input capacitance and stray capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
AA1	1PA1	Input capacitance	23	10	2	15	pF	27
AA2	1PB1	"	3	"	"	"	"	"
AA3	1PC1	"	22	"	"	"	"	"
AA4	1PD1	"	2	"	"	"	"	"
AA5	2PA2	"	7	"	"	"	"	"
AA6	2PB2	"	27	"	"	"	"	"
AA7	2PC2	"	6	"	"	"	"	"
AA8	2PD2	"	26	"	"	"	"	"
BB1	1PW1	Stray capacitance	25	"	0	100	"	"
BB2	1PW2	"	36	"	"	100	"	"
BB3	2PW1	"	30	"	"	50	"	"
BB4	2PW2	"	33	"	"	50	"	"
BB5	2PW1 to 2PW2	"	33	30	5	30	"	"

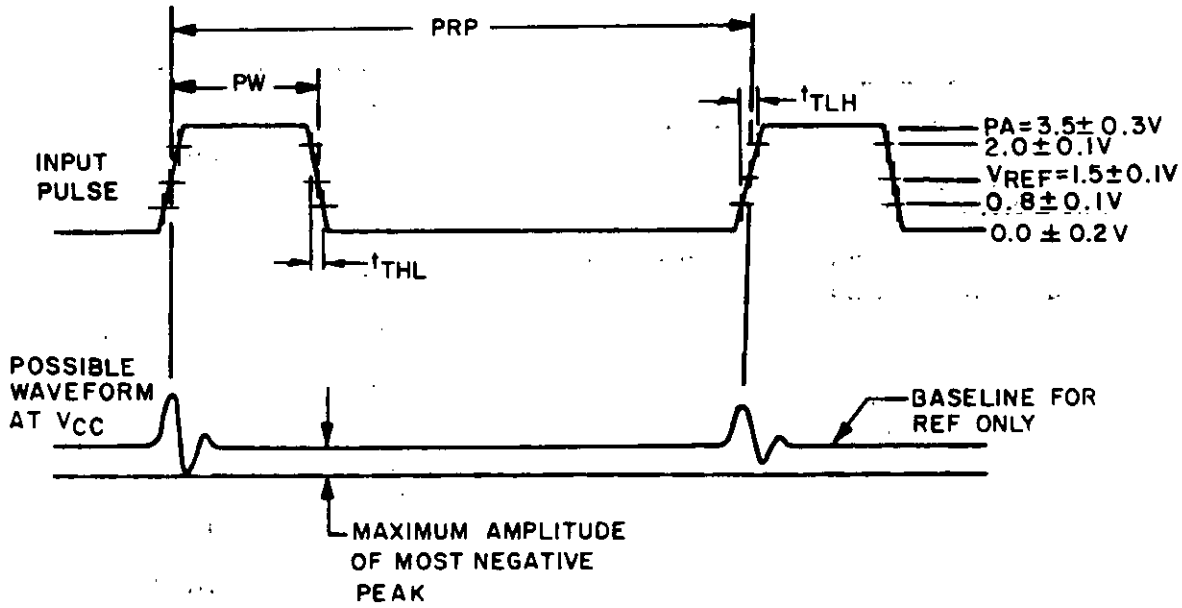
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 25. Power supply transient amplitude test circuit for table XVII.

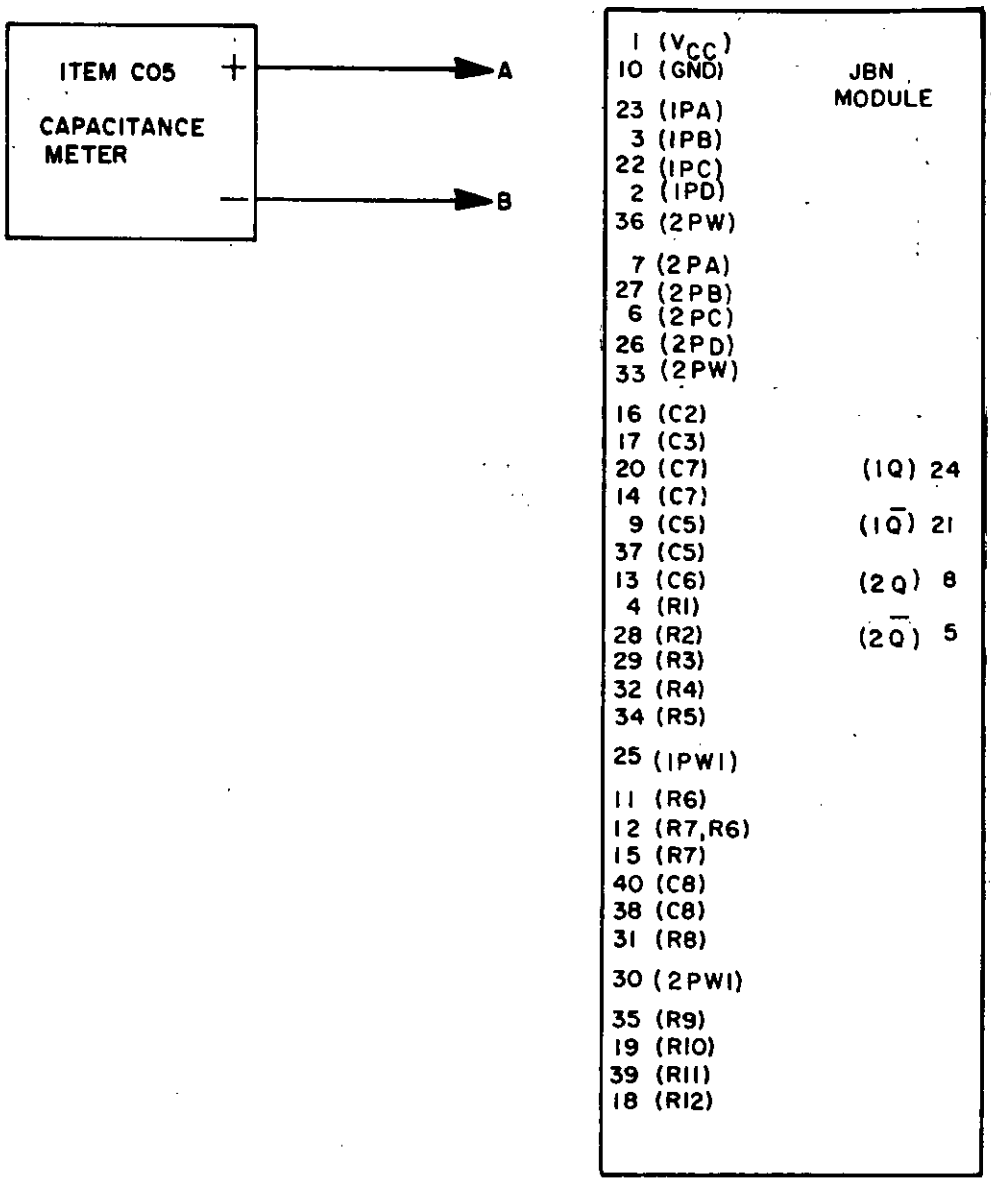
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NOTE: Input pulse parameters:
PW = 200 ±20 ns.
PRP = 800 ±80 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

*FIGURE 26. Power supply transient amplitude test circuit waveforms for table XVII.

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* FIGURE 27. Input capacitance and stray capacitance test circuit for table XVIII.

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